Causal Mediation in Natural Experiments

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Abstract

Natural experiments are a cornerstone of applied economics, providing settings for estimating causal effects with a compelling argument for treatment randomisation, but give little indication of the mechanisms behind causal effects. Causal Mediation (CM) is a framework for sufficiently identifying a mechanism behind the treatment effect, decomposing it into an indirect effect channel through a mediator mechanism and a remaining direct effect. By contrast, a suggestive analysis of mechanisms gives necessary but not sufficient evidence. Conventional CM methods require that the relevant mediator mechanism is as-good-as-randomly assigned; when people choose the mediator based on costs and benefits (whether to visit a doctor, to attend university, etc.), this assumption fails and conventional CM analyses are at risk of bias. I propose an alternative strategy that delivers unbiased estimates of CM effects despite unobserved selection, using instrumental variation in mediator take-up costs. The method identifies CM effects via the Mediator Marginal Treatment Effect, with semi-parametric estimation that is simple to implement in two stages. Applying these methods to the Oregon Health Insurance Experiment reveals a substantial portion of the Medicaid lottery's effect on subjective health and well-being flows through increased healthcare usage — an effect that a conventional CM analysis would mistake. This approach gives applied researchers an alternative method to estimate CM effects when an initial treatment is quasi-randomly assigned, but a mediator mechanism is not, as is common in natural experiments.

Keywords: Direct/indirect effects, quasi-experiment, selection, MTEs.
JEL Codes: C21, C31.

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1 Introduction

Economists use natural experiments to credibly answer social questions, when an experiment was infeasible. For example, does winning access to health insurance causally improve health and well-being (Finkelstein, Taubman, Wright, Bernstein, Gruber, Newhouse, Allen, Baicker & Group 2012)? Natural experiments provide settings which can answer these questions, but give little indication of how these effects came about. Causal Mediation (CM) aims to estimate the mechanisms behind causal effects, by estimating how much of the treatment effect operates through a proposed mediator mechanism. For example, do causal gains from winning access to health insurance come mostly from physical use of healthcare, or plausible psychological gains from no longer having to worry about being uninsured? This study of mechanisms behind causal effects broadens the economic understanding of social settings studied with natural experiments. This paper shows that the conventional approach to estimating CM effects is inappropriate in a natural experiment setting, provides a theoretical framework for how bias operates, and develops an approach to correctly estimate CM effects under alternative assumptions. These methods contrast the current practice in applied economics of providing suggestive evidence of mechanisms, which gives necessary but not sufficient conditions for the mechanisms behind a treatment effect.

In other disciplines — particularly epidemiology, psychology, and medicine — CM has become a standard empirical framework for decomposing causal effects into direct and indirect components (Imai, Keele & Yamamoto 2010). I refer to this established approach as "conventional CM," which assumes both the treatment and the mediator mechanism are (quasi-)randomly assigned. Applied economics has not typically adopted conventional CM, because mediators in economic applications — such as schooling, labour supply, or healthcare use — are choice variables decided by individuals' costs and benefits, likely violating this assumption. Nevertheless, economists often pursue the same goal informally: they present descriptive and suggestive evidence on plausible mediating mechanisms, or occasionally test whether a treatment effect remains after controlling for a plausible mediator mechanism. The first approach gives necessary but not sufficient evidence on the mediating mechanisms, and the second is a conventional CM analysis — despite rarely being named as such by economists.

This paper starts by considering how conventional CM methods (whether applied explicitly or implicitly) perform in a natural experiment setting. Conventional CM methods rely on assuming the initial treatment, and the subsequent mediator mechanism, are both quasi-randomly assigned (Imai et al. 2010). Assuming the mediator is as-good-as-randomly assigned requires either (1) selection is fully captured by observed control variables, or (2) decisions

are effectively random. While these assumptions can be plausible when a mediator is directly manipulated, or the data include extremely rich control variables, they are not credible in most economic applications; when take-up decisions reflect unobserved costs and benefits, mediator assignment is likely not random. For example, in the Oregon Health Insurance Experiment, those who got off the wait-list were randomly chosen by a lottery, but made the choice to visit healthcare in the following year of their own free will; this choice considered their own individual costs and benefits, and thus was not a random choice. I formally derive the selection bias that arises from this non-random assignment, and show through simulations that these biases can be large in settings consistent with standard economic models of selection. This result explains why conventional CM methods can yield misleading conclusions in applied economics, and motivates the need for a framework consistent with quasi-experimental causal reasoning.

Conventional CM's identifying assumptions are at odds with selection based on costs and benefits, so I import methods grounded in labour economic theory to solve the identification problem. This approach identifies CM effects via the Marginal Treatment Effect (MTE) of the mediator, and requires three main assumptions. (1) Mediator take-up must respond only positively to the initial treatment (monotonicity), which implies mediator selection follows a selection model. (2) Mediator take-up is motivated by mediator benefits. (3) A valid instrument for mediator take-up must exist, to avoid relying on parametric assumptions on unobserved selection. While these assumptions are strong, they are plausible in many applied settings. Mediator monotonicity aligns with conventional theories for selection-into-treatment, and is accepted widely in many applications using an instrumental variables research design. Selection based on costs and benefits is central to economic theory, and is the dominant concern for judging observational designs that identify causal effects. Access to valid instrumental variation is a strong condition, though is important to avoid further modelling assumptions; the most compelling example is using variation in mediator take-up costs as an instrument.

Applying the new methods to the Oregon Health Insurance Experiment shows that unobserved selection matters in an analysis of a real-world natural experiment. A substantial portion of the wait-list lottery's impact on subjective health and well-being is mediated indirectly through extra healthcare usage, after instrumenting for healthcare usage with respondents' usual provider. A conventional CM analysis would put this indirect mediated share at practically zero, so that my methods expose that negative selection into healthcare usage would be hiding evidence for this mechanism. These estimates give sufficient evidence that extra healthcare use mediates a sizeable share of the Medicaid-lottery's benefits, though wider confidence intervals underscore the inherent uncertainty on the proportion of the effect

operating through the single mediator mechanism of healthcare usage.

The methods I propose for CM analyses are not perfect for every setting: the structural assumptions are strong, and are tailored to selection-into-mediator based on the economic principle of selection based on costs and benefits. Indeed, this approach provides no safe harbour for estimating CM effects if these structural assumptions do not hold true. This approach imports insights from the instrumental variables literature, connecting the conventional Imai et al. (2010) approach to CM with the economics literature on selection-into-treatment and MTEs (Vytlacil 2002, Heckman & Navarro-Lozano 2004, Heckman & Vytlacil 2005, Florens, Heckman, Meghir & Vytlacil 2008, Brinch, Mogstad & Wiswall 2017, Kline & Walters 2019).

Applied economists mostly investigate mediating mechanisms for causal effects with suggestive analyses of mechanisms. These descriptive analyses are informative, but do not generally identify a mediating pathway without additional assumptions — see also Blackwell, Ma & Opacic (2024), Green, Ha & Bullock (2010). A new strand of the econometric literature has arisen in implicit acknowledgement that suggestive evidence of mechanisms, or a conventional approach to CM, can lead to biased inference and needs alternative methods for credible inference. These include identifying CM effects with overlapping quasi-experimental research designs (Deuchert, Huber & Schelker 2019, Frölich & Huber 2017), functional form restrictions (Heckman & Pinto 2015, Heckman, Pinto & Savelyev 2013), partial identification (Flores & Flores-Lagunes 2009), or an hypothesis test of full mediation through observed channels (Kwon & Roth 2024) — see Huber (2020) for an overview.[1]

I develop a framework showing exactly how selection bias contaminates conventional CM estimates when mediator choices are driven by unobserved gains — settings where none of the existing econometric approaches to CM directly apply. The recent econometric literature has made important progress in adapting CM to observational settings; my work builds on this literature by addressing the common case in which the mediator is not quasi-randomly assigned, when selection into a mediator mechanism aligns with economic theory for selection. Frölich & Huber (2017) is the most similar paper to mine, though I extend the framework for CM from an economic perspective in multiple ways. First, by linking CM to the MTE literature for a binary mediator, I identify average CM effects rather than complier-specific effects (and provide a different estimation approach). Second, I show formally how unobserved selection biases conventional CM estimates and propose an alternative identification strategy that retains the natural experiment structure while relaxing mediator ignorability. Last, this paper provides a rigorous warning to applied economists against uncritically importing

[1] An alternative method to estimate CM effects is ensuring treatment and mediator quasi-random assignment holds by a running two randomised controlled trials for both treatment and mediator, at the same time. This set-up has been considered in the literature previously, in theory (Imai, Tingley & Yamamoto 2013) and in practice (Ludwig, Kling & Mullainathan 2011).

conventional CM methods to avoid the selection bias terms derived, and clarifies the conditions and practices to avoid them.

This paper proceeds as follows. Section 2 describes the dominant approach in economics for studying mechanisms behind treatment effects, illustrating with data from the Oregon Health Insurance Experiment. Section 3 introduces the formal framework for CM, and develops expressions for bias in conventional CM estimates in natural experiment and observational settings. Section 4 describes this bias in applied settings with (1) a regression framework, (2) a setting with selection based on costs and benefits. Section 5 purges bias from CM estimates by identifying CM effects via an MTE approach. Section 6 demonstrates how to estimate CM effects with this approach, with semi-parametric methods, and gives simulation evidence. Section 7 returns to the Oregon Health Insurance Experiment, providing credible estimates of effects on subjective health and well-being mediated through increased healthcare usage. Section 8 concludes.

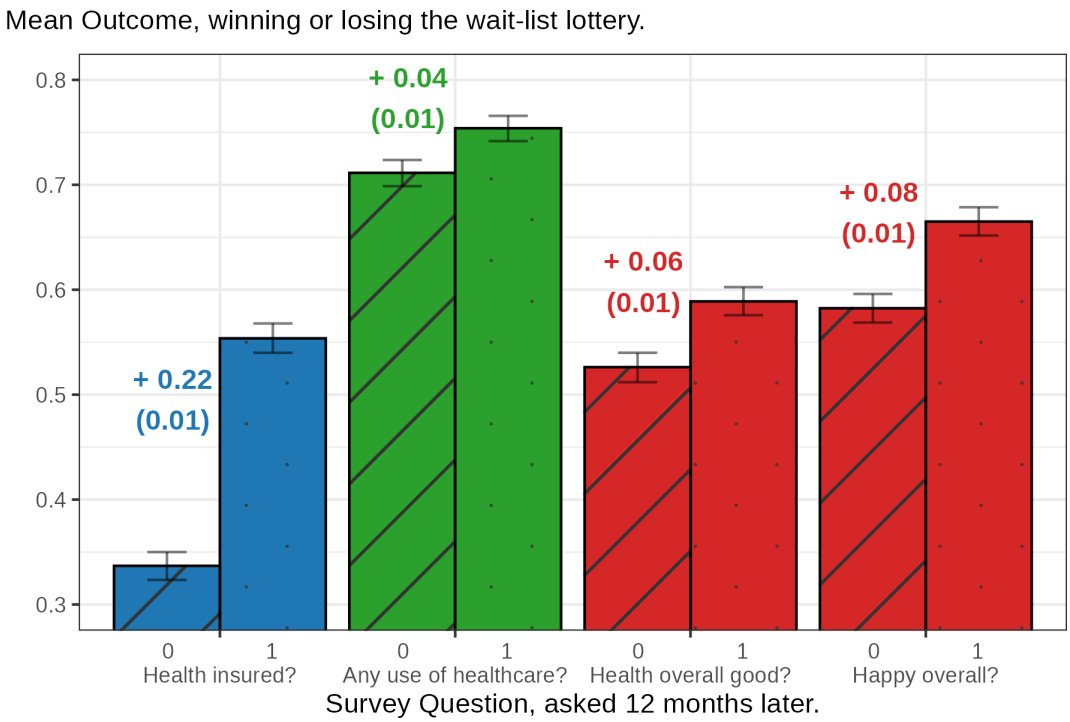
2 Mechanisms in the Oregon Health Insurance Experiment

In the United States, healthcare is generally not provided directly by the government. Instead, consumers purchase health insurance to fund healthcare expenses, with the government providing insurance only for elderly individuals (Medicare) and for those with low-incomes (Medicaid). In 2004, the state of Oregon ceased accepting new applications for Medicaid due to budgetary constraints, and did not reopen applications until 2008. When the state resumed enrolment, 90,000 individuals applied, vastly exceeding the programme's capacity. Oregon therefore allocated the opportunity to apply for Medicaid via a lottery system among those on the wait-list. Winning this wait-list lottery significantly increased healthcare usage, plus subjective health and well-being.

Winning the wait-list lottery increased the average health insurance coverage rate by 22 percentage points (pp), and subjective visitation of any healthcare provider in the following 12 months by 4 pp. In addition, wait-list lottery winners agreed 6 pp more with the question "In general, would you say your health is excellent, very good, or good" (hereafter, subjective health), and 8 pp for "How would you say things are these days-would you say that you are very or pretty happy" (hereafter, subjective well-being). These numbers are calculated among the 9,957 people eligible for the Oregon wait-list lottery who responded to a survey sent by Finkelstein et al. (2012).[2] Figure 1 summarises these results.

[2]This number restricts to those who gave non-missing answers to all relevant questions, including ques-

Figure 1: Effects of the Oregon Health Insurance Experiment Wait-list Lottery.



Note: This figure summarises the relevant results of the Oregon Health Insurance Experiment (Finkelstein et al. 2012). $\mathbb{E}[Y_i \mid Z_i = z]$ is the mean outcome, where $z = 0$ refers to the case of losing the wait-list lottery (not given access to Medicaid) and $z = 1$ winning. The numbers beside the bars are estimates of the mean difference; winning the Medicaid wait-list lottery increased average health insurance rate by 22 percentage points (pp), with standard errors of the difference reported in brackets.

These results show that winning the wait-list lottery led to large gains in subjective health and well-being. The economics, medicine, and health policy literatures have primarily focused on the health benefits — often interpreted as healthcare benefits from new access to government provided health insurance.[3] However, the original authors also noted other benefits, including complete elimination of catastrophic out-of-pocket medical debt among those with new access to Medicaid. These are plausibly income effects that benefit recipients directly, not only through increased use of healthcare, but also by reducing stress and improving financial security. These plausible direct effects have not been explored in the applied literature.

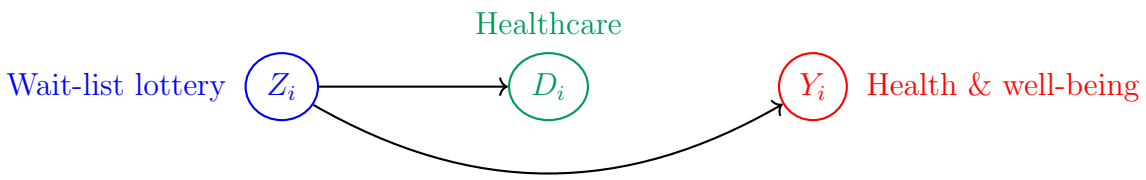
Accepted practice in applied economics is to investigate mechanisms behind causal effects with suggestive evidence. This involves estimating the average causal effect of the wait-lottery

tions on pre-lottery location of usual healthcare, using anonymised data from the Oregon Health Insurance Experiment replication package (Finkelstein & Baicker 2014).

[3] Finkelstein et al. (2012) use the wait-list lottery as an IV because health insurance is not randomly assigned; this paper focuses on the average effects of winning the wait-list lottery (which is randomly assigned).

on a proposed mediator (healthcare usage) and separately estimating its effect on the final outcomes (subjective health and well-being). When both estimates are positive, and the mediator precedes the outcome, it is taken as de facto evidence that the mediator transmits the treatment effect. In the case of the Oregon Health Insurance Experiment, this amounts to concluding that increased healthcare usage mediates the positive effects of winning the lottery on health and well-being. Figure 2 illustrates this approach, which is also prevalent in other social science fields — see Blackwell et al. (2024), Green et al. (2010).

Figure 2: Structural Causal Model for Suggestive Evidence of a Mechanism.



Note: This figure shows the structural causal model behind a suggestive analysis for effects of the Oregon Health Insurance Experiment, where arrows represent causal effects — e.g., $Z_i \to D_i$ means Z_i affects D_i with no reverse causality.

This approach gives necessary, but not sufficient, identification of healthcare as a mediating mechanism. It is not sufficient because it provides no evidence for the effect of healthcare on health and well-being, so does not identify the causal mechanism. Studying this mechanism with suggestive evidence require an additional, hidden assumption that healthcare positively affects health outcomes. While this assumption is not unreasonable in general, it may not apply within the data under study; subjective well-being is measured only 12 months after the Medicaid lottery, which may not be enough time for health gains to accrue. Second, this approach does not quantify the mechanism effects. Healthcare could have a very large effect on subjective health and well-being, or possibly a very small effect — it is a priori unclear. In addition, the mediator mechanism effect refers not to the average effect of healthcare usage, but to the effect for Oregon residents who were induced to use more healthcare after winning the wait-list lottery (mediator compliers). This local effect could differ substantially from a population average, and potentially mislead conclusions about the magnitude or generality of the mechanism. Together, these concerns mean that a suggestive analysis of healthcare as a mediating mechanism are not dispositive; without additional assumptions, they neither identify nor quantify the mediator mechanism channel.

CM offers a compelling alternative framework, explicitly defining the average direct and indirect effects and clear assumptions under which they are identified. Moreover, it delivers quantitative answers to the key question: how much of a treatment effect operates through a

specific mediator mechanism? CM is widely used in fields such as epidemiology, psychology, and medicine where researchers regularly decompose treatment effects into component pathways. However, CM methods have not yet been examined from an economic perspective to assess their applicability in observational causal research, such as natural experiments.

3 Causal Mediation (CM)

CM decomposes causal effects into two channels, through a mediating mechanism (indirect effect) and through all other paths (direct effect). To develop notation, write $Z_i = 0, 1$ for a binary treatment, $D_i = 0, 1$ a binary mediator mechanism, and Y_i a continuous outcome for individuals $i = 1, \ldots, n$.[4] D_i and Y_i are a sum of their potential outcomes,

$$\begin{aligned} D_i &= (1 - Z_i)D_i(0) + Z_i D_i(1), \\ Y_i &= (1 - Z_i)Y_i(0, D_i(0)) + Z_i Y_i(1, D_i(1)). \end{aligned}$$

Assume treatment Z_i is quasi-randomly assigned,[5]

$$Z_i \perp\!\!\!\perp D_i(z), Y_i(z', d), \text{ for } z, z', d = 0, 1.$$

There are only two average effects which are identified without additional assumptions.

1. The average first-stage refers to the effect of the treatment on mediator, Z_i on D_i:

$$\mathbb{E}\left[D_i \,|\, Z_i = 1\right] - \mathbb{E}\left[D_i \,|\, Z_i = 0\right] = \Pr\left(D_i(0) \neq D_i(1)\right).$$

It is common in the economics literature to also assume that Z_i influences D_i in at most one direction, $D_i(0) \leq D_i(1)$ almost surely — monotonicity (Imbens & Angrist 1994). I assume mediator monotonicity holds throughout, to simplify notation.

2. The Average Treatment Effect (ATE) refers to the effect of the treatment on outcome, Z_i on Y_i, and is also known as the average total effect or intent-to-treat effect in social science settings, or reduced-form effect in the instrumental variables literature:

$$\mathbb{E}\left[Y_i \,|\, Z_i = 1\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0\right] = \mathbb{E}\left[Y_i(1, D_i(1)) - Y_i(0, D_i(0))\right].$$

[4] This paper exclusively focuses on the binary case. See Huber, Hsu, Lee & Lettry (2020) or Frölich & Huber (2017) for a discussion of CM with continuous treatment and/or mediator, and the assumptions required.

[5] This assumption can hold conditional on a covariate vector, $\boldsymbol{X}_i$. To simplify notation in this section, leave the conditional part unsaid, as it changes no part of the identification framework.

Z_i affects outcome Y_i directly, and indirectly via the $D_i(Z_i)$ channel, with no reverse causality. CM aims to decompose the ATE of Z_i on Y_i into these two separate pathways. For each $z = 0, 1$,

$$\underbrace{\mathbb{E}\left[Y_i(1, D_i(1)) - Y_i(0, D_i(0))\right]}_{\text{ATE}} = \underbrace{\mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0))\right]}_{\text{Average natural Indirect Effect, AIE}(z).} + \underbrace{\mathbb{E}\left[Y_i(1, D_i(1-z)) - Y_i(0, D_i(1-z))\right]}_{\text{Average natural Direct Effect, ADE}(1-z).}$$

These parameters are conventionally termed the average natural direct and indirect effects; for brevity, I refer to them henceforth as the ADE and AIE, respectively.[6] The index on each effect records the channel held fixed when separating the two causal pathways. AIE(z) changes the mediator from $D_i(0)$ to $D_i(1)$ while holding treatment fixed at the relevant treatment state $z = 0, 1$; ADE(z) changes treatment from $Z_i = 0$ to $Z_i = 1$ while holding the mediator choice at $D_i(1-z)$ for the opposite treatment state $1-z$. This distinction matters when the effects of Z_i and D_i interact: the effect of changing one may depend on the level at which the other is held fixed. ATE = ADE(1) + AIE(0) has the clearest interpretation as the effect of changing the mediator from $D_i(0)$ to $D_i(1)$ while holding treatment at the untreated baseline $Z_i = 0$, then allocating the remainder to the direct effect.[7]

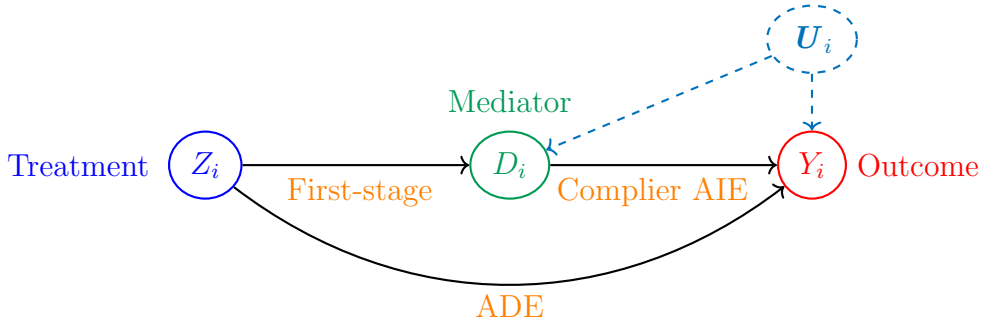
Estimating the AIE answers the following question: how much of the causal effect Z_i on Y_i goes through the D_i channel? Figure 3 visualises the design, where the direction arrows denote the causal direction. When studying the health gains of winning the Medicaid wait-list lottery (Finkelstein et al. 2012), the AIE represents how much of the effect comes from using the hospital more often. Estimating the ADE answers the following equation: how much is left over after accounting for the D_i channel? For the example, how much of the wait-list lottery effect is a direct effect, other than increased healthcare usage — e.g., income effects of lower medical debt, or less worry over health shocks thanks to government support. An Instrumental Variables (IV) approach assumes this direct effect is zero for everyone (the exclusion restriction). CM is a similar, yet distinct, framework attempting to explicitly model the direct effect, and not assuming it is zero.

The ADE and AIE are not separately identified without further assumptions.

[6] Terminology for CM is somewhat fractured across the literature. I follow the terminology favoured by Huber (2020). Imai et al. (2010) labels AIE(z) the Average Causal Mediation Effect, and ADE(z) the Average Direct Effect; Pearl (2003) labels them the natural indirect effect and natural direct effect, respectively; and Robins (2003) labels AIE(0) and ADE(0) the pure indirect and total direct effects, respectively, and AIE(1) and ADE(1) the total indirect and pure direct effects, respectively.

[7] All arguments developed below apply symmetrically to the alternative decomposition, ATE = ADE(0) + AIE(1).

Figure 3: Structural Causal Model for CM.



Note: This figure shows the structural causal model behind CM. The Complier AIE refers to the AIE local to $D_i(Z_i)$ compliers, the group with $D_i(0) = 0, D_i(1) = 1$, so that AIE = Average First-stage × Complier AIE. $\boldsymbol{U}_i$ shows an unobserved confounding variable for the causal effect $D_i \to Y_i$, representing this paper's focus on the case that D_i is not quasi-randomly assigned. Subsection 4.1 defines $\boldsymbol{U}_i$ in an applied setting.

3.1 Identification of CM Effects

The conventional approach to estimating direct and indirect effects assumes both Z_i and D_i are quasi-randomly assigned, conditional on a vector of control variables $\boldsymbol{X}_i$.

Definition 1. *Sequential quasi-random assignment (Imai et al. 2010)*

$$Z_i \perp\!\!\!\perp D_i(z), Y_i(z', d) \mid \boldsymbol{X}_i, \qquad \text{for } z, z', d = 0, 1 \tag{1}$$

$$D_i \perp\!\!\!\perp Y_i(z, d) \mid \boldsymbol{X}_i, Z_i = z, \qquad \text{for } z, d = 0, 1. \tag{2}$$

Sequential quasi-random assignment assumes that the initial treatment Z_i is quasi-randomly assigned conditional on $\boldsymbol{X}_i$ (as has already been assumed above). It then also assumes that, after Z_i is assigned, that D_i is quasi-randomly assigned conditional on $\boldsymbol{X}, Z_i$ (hereafter, mediator quasi-random assignment). If 1(1) and 1(2) hold, then the ADE and AIE are identified by two-stage mean differences conditioning on $\boldsymbol{X}_i$.[8]

[8] In addition, a common support condition for both Z_i, D_i (across $\boldsymbol{X}_i$) is necessary. Imai et al. (2010) shows this general identification statement; see Appendix A.1.

$$\mathbb{E}_{\boldsymbol{X}_i}\left[\mathbb{E}_{D_i \mid Z_i=z,\boldsymbol{X}_i}\left[\underbrace{\mathbb{E}\left[Y_i \mid Z_i=1, D_i, \boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i \mid Z_i=0, D_i, \boldsymbol{X}_i\right]}_{\text{Second-stage mean difference, } Z_i \to Y_i \text{ holding } D_i \text{ constant to its distribution under treatment state } Z_i=z}\right]\right] = \underbrace{\mathbb{E}\left[Y_i(1, D_i(z))-Y_i(0, D_i(z))\right]}_{\text{ADE}(z), \text{ for } z=0,1.}$$

$$\mathbb{E}_{\boldsymbol{X}_i}\left[\underbrace{\left(\mathbb{E}\left[D_i \mid Z_i=1, \boldsymbol{X}_i\right]-\mathbb{E}\left[D_i \mid Z_i=0, \boldsymbol{X}_i\right]\right)}_{\text{First-stage mean difference, } Z_i \to D_i} \times \underbrace{\left(\mathbb{E}\left[Y_i \mid Z_i=z, D_i=1, \boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i \mid Z_i=z, D_i=0, \boldsymbol{X}_i\right]\right)}_{\text{Second-stage mean difference, } D_i \to Y_i \text{ holding } Z_i \text{ constant at } z}\right]$$
$$= \underbrace{\mathbb{E}\left[Y_i(z, D_i(1))-Y_i(z, D_i(0))\right]}_{\text{AIE}(z), \text{ for } z=0,1.}$$

I refer to the estimands on the left-hand side as conventional CM estimands, which are typically estimated by a composition of two-stage Ordinary Least Squares (OLS) estimates (Imai et al. 2010). While this is the most common approach in the applied literature, I do not assume the linear model for my identification analysis. Linearity assumptions are not necessary for identification, and it suffices to note that heterogeneous treatment effects and non-linear confounding can bias OLS estimates of CM estimands in the same manner that is well documented elsewhere (see e.g., Angrist 1998, Słoczyński 2022). This section focuses on problems that plague conventional CM, regardless of estimation method.

3.2 Non-identification of CM Effects

Applied research often uses a natural experiment to study settings where treatment Z_i is quasi-randomly assigned, justifying assumption 1(1). Rarely do they also have access to an additional, overlapping natural experiment to isolate random variation in D_i — to justify mediator quasi-random assignment 1(2). One might consider conventional CM methods in such a setting to learn about the mechanisms behind the causal effect Z_i on Y_i. This approach leads to estimates at risk of bias, contaminating inference on direct and indirect effects.

Theorem 1. *Absent an identification strategy for the mediator, conventional CM estimates are at risk of selection bias. If 1(1) holds, and 1(2) does not, then conventional CM estimands are contaminated by selection bias and group differences. Proof: see Appendix A.2.*

Below I present the relevant selection bias and group difference terms, omitting the conditional on $\boldsymbol{X}_i$ notation for brevity.

For the ADE: conventional CM estimand = ADE + selection bias + group differences.[9]

$$\begin{aligned}
&\mathbb{E}_{D_i \mid Z_i=z}\Big[\mathbb{E}\left[Y_i \mid Z_i=1, D_i\right] - \mathbb{E}\left[Y_i \mid Z_i=0, D_i\right]\Big] \\
&= \mathrm{ADE}(z) \\
&\quad + \mathbb{E}_{D_i=d \mid Z_i=z}\Big[\mathbb{E}\left[Y_i(0,d) \mid D_i(1)=d\right] - \mathbb{E}\left[Y_i(0,d) \mid D_i(0)=d\right]\Big] \\
&\quad + \mathbb{E}_{D_i=d \mid Z_i=z}\Big[\mathbb{E}\left[Y_i(1,d)-Y_i(0,d) \mid D_i(1)=d\right] - \mathbb{E}\left[Y_i(1,d)-Y_i(0,d) \mid D_i(z)=d\right]\Big], \text{ for } z=0,1.
\end{aligned}$$

For the AIE: conventional CM estimand = AIE + selection bias + group differences.

$$\begin{aligned}
&\Big(\mathbb{E}\left[D_i \mid Z_i=1\right] - \mathbb{E}\left[D_i \mid Z_i=0\right]\Big) \times \Big(\mathbb{E}\left[Y_i \mid Z_i=z, D_i=1\right] - \mathbb{E}\left[Y_i \mid Z_i=z, D_i=0\right]\Big) \\
&= \mathrm{AIE}(z) \\
&\quad + \Pr\left(D_i(0)=0, D_i(1)=1\right)\Big(\mathbb{E}\left[Y_i(z,0) \mid D_i(z)=1\right] - \mathbb{E}\left[Y_i(z,0) \mid D_i(z)=0\right]\Big) \\
&\quad + \Pr\left(D_i(0)=0, D_i(1)=1\right)\begin{pmatrix}\mathbb{E}\left[Y_i(z,1)-Y_i(z,0) \mid D_i(z)=1\right] \\ -\mathbb{E}\left[Y_i(z,1)-Y_i(z,0) \mid D_i(0)=0, D_i(1)=1\right]\end{pmatrix}, \text{ for } z=0,1.
\end{aligned}$$

The selection bias terms come from systematic differences between the groups taking or refusing the mediator ($D_i(z)=1$ versus $D_i(z)=0$), differences not fully unexplained by $\boldsymbol{X}_i$. These selection bias terms would equal zero if the mediator had been quasi-randomly assigned 1(2), but do not necessarily average to zero if not. In the Oregon Health Insurance Experiment, the wait-list gave random variation in the treatment (the Medicaid wait-list lottery) but there was not a similar natural experiment for healthcare usage; correspondingly, the selection-on-observables approach to CM has selection bias.

The group differences represent the fact that mean differences give an average effect on the $D_i(z)=1$ treated group, which is systematically different from the compliers' average effect if selection-on-observables does not hold. These terms are a non-parametric framing of the bias from controlling for intermediate outcomes, previously studied only in a linear setting (i.e., bad controls in Cinelli, Forney & Pearl 2024, or M-bias in Ding & Miratrix 2015).

The AIE group differences term is longer, because the indirect effect is the average effect

[9] The bias terms here mirror those in Heckman, Ichimura, Smith & Todd (1998), Angrist & Pischke (2009) for a single $D_i \to Y_i$ average causal effect, when D_i is not quasi-randomly assigned:

$$\mathbb{E}\left[Y_i \mid D_i=1\right] - \mathbb{E}\left[Y_i \mid D_i=0\right] = \mathrm{ATE} + \underbrace{\Big(\mathbb{E}\left[Y_i(.,0) \mid D_i=1\right] - \mathbb{E}\left[Y_i(.,0) \mid D_i=0\right]\Big)}_{\text{Selection Bias}} + \underbrace{\Pr\left(D_i=0\right)\left(\mathrm{ATT}-\mathrm{ATU}\right)}_{\text{Group-differences Bias}}.$$

of D_i local to $D_i(Z_i)$ compliers.

$$\begin{aligned}\text{AIE}(z) &= \mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0))\right] \\ &= \Pr\left(D_i(0) = 0, D_i(1) = 1\right) \underbrace{\mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \,|\, D_i(0) = 0, D_i(1) = 1\right]}_{\text{Average } D_i \to Y_i \text{ effect among } D_i(Z_i) \text{ compliers}}\end{aligned}$$

It is important to acknowledge the mediator compliers here, because the AIE is the treatment effect going through the $D_i(Z_i)$ channel, thus only refers to individuals pushed into mediator D_i by initial treatment Z_i.[10] If we had been using a population average effect for D_i on Y_i, then this is losing focus on the definition of the AIE; it is not about the causal effect D_i on Y_i, it is about the causal effect $D_i(Z_i)$ on Y_i.

The group difference bias term arises because the convention approach to CM assumes the complier average effect is equal to the population average effect, which does not hold true if the mediator is not quasi-randomly assigned.

4 CM in Applied Settings

Unobserved confounding is particularly problematic when studying the mechanisms behind treatment effects. For example, in studying health gains from the Oregon wait-list lottery, we might expect that health gains came about because those who won access to Medicaid started visiting their healthcare provider more often, when in past they avoided it over financial concerns. Applying conventional CM methods to investigate this expectation would be dismissing unobserved confounders for how often individuals visit healthcare providers, leading to biased results.

The wider population does not have one uniform bill of health; many people are born predisposed to ailments, due to genetic variation or other unrelated factors. These conditions can exist for years before being diagnosed. People with severe underlying conditions may visit healthcare providers more often than the rest of the population, to investigate or begin treating the ill-effects. It stands to reason that people with more serve underlying conditions may gain more from more often attending healthcare providers once given health insurance. These underlying causes cannot be controlled for by researchers, as we cannot hope to observe and control for health conditions that are yet to even be diagnosed. This means underlying health conditions are an unobserved confounder, and will bias estimates of the ADE and AIE in this setting.

In this section, I further develop the issue of selection on unobserved factors in a general

[10] This section has relied on mediator monotonicity, so that the mediator defier share is almost surely zero, $\Pr\left(D_i(0) = 0, D_i(1) = 1\right) = 0$. Appendix A.3 describes the AIE (and selection bias) without monotonicty.

CM setting. First, I show the non-parametric bias terms from Section 3 can be written as omitted variables bias in a random coefficients regression framework. Second, I show how selection bias operates in a basic model for selection-into-mediator based on costs and benefits.

4.1 Regression Framework

Inference for CM effects can be written in a regression framework with random coefficients, showing how correlation between unobserved error terms and the mediator disrupts identification.

Start by writing potential outcomes $Y_i(.,.)$ as a sum of observed and unobserved factors, following the notation of Heckman & Vytlacil (2005). For each $z, d = 0, 1$, put $\mu_d(z; \boldsymbol{X}_i) = \mathbb{E}\left[Y_i(z,d) \mid \boldsymbol{X}_i\right]$ and the corresponding error terms, $U_i(z,d) = Y_i(z,d) - \mu_d(z; \boldsymbol{X}_i)$, so we have the following expressions:

$$Y_i(Z_i, 0) = \mu_0(Z_i; \boldsymbol{X}_i) + U_i(Z_i, 0), \quad Y_i(Z_i, 1) = \mu_1(Z_i; \boldsymbol{X}_i) + U_i(Z_i, 1).$$

With this notation, observed data $Z_i, D_i, Y_i, \boldsymbol{X}_i$ have the following random coefficient outcome formulae — which characterise direct effects, indirect effects, and selection bias.[11]

$$D_i = \theta + \overline{\pi} Z_i + \zeta(\boldsymbol{X}_i) + \eta_i, \tag{3}$$

$$Y_i = \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\boldsymbol{X}_i) + \varepsilon_i. \tag{4}$$

First-stage (3) is identified, with $\theta + \zeta(\boldsymbol{X}_i)$ the intercept, and $\overline{\pi}$ the first-stage average compliance rate (conditional on $\boldsymbol{X}_i$). Second-stage (4) has the following definitions, and is not identified thanks to the correlated error term, ε_i.

(a) $\alpha = \mathbb{E}\left[\mu_0(0; \boldsymbol{X}_i)\right]$ and $\varphi(\boldsymbol{X}_i) = \mu_0(0; \boldsymbol{X}_i) - \alpha$ are the intercept terms.

(b) $\beta = \mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)$ is the controlled indirect effect under $Z_i = 0$ conditional on $\boldsymbol{X}_i$.

(c) $\gamma = \mu_0(1; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)$ is the controlled direct effect under $D_i = 0$ conditional on $\boldsymbol{X}_i$.

(d) $\delta = \mu_1(1; \boldsymbol{X}_i) - \mu_0(1; \boldsymbol{X}_i) - \big(\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)\big)$ is the average interaction effect conditional on $\boldsymbol{X}_i$.

[11] This is not consequence of linearity assumptions; the outcome formulae allow for unconstrained heterogeneous treatment effects, because the coefficients are random. If either Z_i, D_i were continuously distributed, then this representative would not necessarily hold true.

(e) ε_i is the disruptive error term, with the following definition

$$\varepsilon_i = \begin{cases} (1 - D_i)\, U_i(0,0) + D_i U_i(0,1), & \text{if } Z_i = 0 \\ (1 - D_i)\, U_i(1,0) + D_i U_i(1,1), & \text{if } Z_i = 1. \end{cases}$$

The selection bias comes from the fact that ε_i is correlated with mediator D_i, when D_i is not quasi-randomly assigned.

The ADE and AIE are averages of these random coefficients:

$$\text{ADE}(z) = \mathbb{E}\left[\gamma + \delta D_i \,|\, Z_i = z\right], \qquad \text{AIE}(z) = \mathbb{E}\left[\overline{\pi}\Big(\beta + \delta z + \widetilde{U}_i(z)\Big)\right],$$

where $\widetilde{U}_i(z) = \mathbb{E}\left[U_i(z,1) - U_i(z,0) \,|\, D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]$, for each $z = 0, 1$. Second-stage (4) is not identified, so that the ADE and AIE are not identified in these regression equations — Appendix A.4 shows the omitted variables bias formulae.

By construction, $\boldsymbol{U}_i \coloneqq \Big(U_i(z,d)\Big)_{(z,d)\in\{0,1\}^2}$ is an unobserved confounder. The regression estimates of β, γ, δ in second-stage (4) give unbiased estimates if D_i is also conditionally quasi-randomly assigned: $D_i \perp\!\!\!\perp \boldsymbol{U}_i$. If not, then estimates of CM effects suffer from omitted variables bias from failing to adjust for the unobserved confounder, $\boldsymbol{U}_i$.

4.2 Selection on Costs and Benefits

CM is at risk of bias because $D_i \perp\!\!\!\perp \boldsymbol{U}_i$ is unlikely to hold in applied settings. A separate identification strategy could disrupt the selection-into-D_i based on unobserved factors, and lend credibility to the mediator quasi-random assignment assumption. Without it, bias will persist, given how we conventionally think of selection-into-treatment.

Consider a model where individual i selects into a mediator based on costs and benefits (in terms of outcome Y_i), after $Z_i, \boldsymbol{X}_i$ have been assigned. In a natural experiment setting, an external factor has disrupted individuals selecting Z_i by choice (thus Z_i is quasi-randomly assigned), but it has not disrupted the choice to take mediator (thus D_i is not quasi-randomly assigned). In the Oregon Health Insurance Experiment, the treatment variation comes from the wait-list lottery, while healthcare usage was not subject to a similar lottery. Write C_i for individual i's costs of taking mediator D_i, and $\mathbb{1}\{.\}$ for the indicator function. The Roy model has i taking the mediator if the benefits exceed the costs,

$$D_i(z) = \mathbb{1}\left\{ \underbrace{C_i}_{\text{Costs}} \;\leq\; \underbrace{Y_i(z,1) - Y_i(z,0)}_{\text{Benefits}} \right\}, \quad \text{for } z = 0, 1. \tag{5}$$

The Roy model provides an intuitive framework for analysing selection mechanisms be-

cause it captures the fundamental economic principle of decision-making based on costs and benefits in terms of the outcome under study (Roy 1951, Heckman & Honore 1990). In the Oregon Health Insurance Experiment, this models choice to visit the doctor in terms of health and well-being benefits relative to costs.[12] This makes the Roy model useful as a base case for CM, where selection-into-mediator may be driven by private information (unobserved by the researcher).

By using the Roy model as a benchmark, I explore the practical limits of the mediator quasi-random assignment assumption. If selection follows a Roy model, and the mediator is quasi-randomly assigned, then unobserved benefits can play no part in selection. The only driver of selection are individuals' differences in costs (and not benefits). If there are any selection-into-D_i benefits unobserved to the researcher, then mediator quasi-random assignment cannot hold.

Proposition 1. *Suppose mediator selection follows a Roy model* (5)*, and selection is not fully explained by costs and observed gains. Then mediator quasi-random assignment does not hold, and thus conventional CM estimands are at risk of selection bias.*

This is an equivalence statement: selection based on costs and benefits is only consistent with mediator quasi-random assignment if the researcher observed every single source of mediator benefits. See Appendix A.5 for the proof. This means than the vector of control variables $\boldsymbol{X}_i$ must be incredibly rich. Together, $\boldsymbol{X}_i$ and unobserved cost differences $U_{C,i}$ must explain selection-into-D_i one hundred percent. In the Roy model framework, however, individuals make decisions about mediator take-up based on gains — whether the researcher observes them or not. The unobserved gains are unlikely to be fully captured by an observed control set $\boldsymbol{X}_i$, except in very special cases.

In practice, the best setting to believe in the mediator quasi-random assignment assumption is to study a setting where the researcher has two causal research designs, one for treatment Z_i and another for mediator D_i, at the same time. Absent this, mediator quasi-random assignment become hard to believe, and the corresponding conventional CM estimates are at risk of selection bias.

5 Solving Identification via the Mediator MTE

If your goal is to estimate CM effects, and you could control for unobserved selection terms $\boldsymbol{U}_i$, then you would. This ideal (but infeasible) scenario would yield unbiased estimates

[12] If the choice is considered over a sum of outcomes, then a simple extension to a utility maximisation model maintains this same framework with expected costs and benefits. See Heckman & Honore (1990), Eisenhauer, Heckman & Vytlacil (2015).

for the ADE and AIE. Identification via the mediator Marginal Treatment Effect (MTE) takes this insight seriously, providing conditions to model the implied confounding and thus avoiding it.

The main problem is that second-stage regression equation (4) is not identified, because $\boldsymbol{U}_i$ is unobserved, and leads to omitted variables bias.

$$\begin{aligned}\mathbb{E}\left[Y_i \mid Z_i, D_i, \boldsymbol{X}_i\right] &= \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\boldsymbol{X}_i) \\ &+ \underbrace{(1-D_i)\,\mathbb{E}\left[U_i(Z_i,0) \mid D_i(Z_i)=0, \boldsymbol{X}_i\right] + D_i \mathbb{E}\left[U_i(Z_i,1) \mid D_i(Z_i)=1, \boldsymbol{X}_i\right]}_{\text{Unobserved confounding.}}\end{aligned} \tag{6}$$

My approach to identifying CM effects models unobserved selection-into-D_i via the mediator MTE, avoiding the bias terms derived in Section 3. Thinking on MTEs began with corrections for sample selection problems (Heckman 1974), and were extended to a switching/selection problem in parametric settings (Heckman 1979, Björklund & Moffitt 1987) and a general case (Heckman & Vytlacil 2005). The approach works in the following manner: (1) assume that the variable of interest follows a threshold-crossing model, where unexplained first-stage selection informs unobserved second-stage confounding; (2) extract information about unobserved confounding from the first-stage using an instrumental variable; and (3) identify the ADE and AIE by interpolating the Mediator MTE.

Identification in MTE methods typically relies on identifying the corresponding Control Functions (CFs) with an external instrument or distributional assumptions; this paper focuses exclusively on the case that an instrument is available. By explicitly accounting for the information contained in the first-stage selection model, the MTE approach enables consistent estimation of causal effects in the second-stage even when selection is driven by unobserved factors.

In the OHIE application, the MTE approach allows unobserved determinants of healthcare use to be associated with potential health outcomes. Instrument-induced variation in healthcare use then reveals how the causal effect of healthcare varies along this latent selection margin. Then it uses this information in the second-stage estimation of how much the effect goes through increased healthcare usage, estimating the ADE and AIE after controlling for this confounding.

5.1 MTE Model Set-up

The mediator MTE measures how the average causal effect of mediator D_i varies across the population distribution, measured by a uniform random variable U_i.

$$\begin{aligned}\text{MTE}(p; z, \boldsymbol{X}_i) &= \mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \mid \boldsymbol{X}_i, U_i = p\right] \\ &= \beta + \delta z + \mathbb{E}\left[U_i(z,1) - U_i(z,0) \mid \boldsymbol{X}_i, U_i = p\right], \qquad \text{for } z = 0,1,\ p \in (0,1).\end{aligned}$$

Under the threshold-crossing selection model, a marginal increase in the propensity score induces mediator take-up among individuals whose latent mediator take-up cost U_i is locally equal to p.

The following assumptions interpolate the mediator MTE across the relevant support, and thus both the ADE and AIE.

Assumption MTE–1. Mediator monotonicity.

$$\Pr\left(D_i(0) \leq D_i(1)\right) = 1.$$

Assumption MTE–1 is the monotonicity condition first used in an IV context (Imbens & Angrist 1994). Here, it is assuming that people respond to treatment, Z_i, by consistently taking or refusing the mediator D_i (always or never-mediators), or taking the mediator D_i if and only if assigned to the treatment $Z_i = 1$ (mediator compliers). There are no mediator defiers, so that the mediator switcher group ($D_i(0) \neq D_i(1)$) is comprised entirely of mediator compliers ($D_i(0) = 0$, $D_i(1) = 1$).

The main implication of Assumption MTE–1 is that selection-into-mediator can be written as a selection model with ordered threshold crossing values that describe selection-into-D_i (Vytlacil 2002).

$$D_i(z) = \mathbb{1}\left\{V_i \leq \psi\big(z; \boldsymbol{X}_i\big)\right\}, \quad \text{for } z = 0,1$$

where V_i is a latent variable with continuous distribution and conditional cumulative density function $F_V(.\,|\boldsymbol{X}_i)$, and $\psi(.\,;\boldsymbol{X}_i)$ collects observed sources of mediator selection. V_i could be assumed to follow a known distribution; the canonical Heckman selection model assumes V_i is normally distributed (a "Heckit" model). The identification strategy here applies to the general case that the distribution of V_i is unknown, without parametric restrictions.

I focus on the equivalent transformed model of Heckman & Vytlacil (2005),

$$D_i(z) = \mathbb{1}\left\{U_i \leq \pi(z; \boldsymbol{X}_i)\right\}, \quad \text{for } z = 0,1$$

where $U_i \coloneqq F_V\left(V_i \mid \boldsymbol{X}_i\right)$ follows a uniform distribution, and $\pi(z; \boldsymbol{X}_i) = F_V\big(\psi(z; \boldsymbol{X}_i)\big) = \Pr\left(D_i = 1 \,|\, Z_i = z, \boldsymbol{X}_i\right)$ is the mediator propensity score. U_i are the unobserved mediator

take-up costs. Note the maintained assumption that treatment Z_i is quasi-randomly assigned conditional on $\boldsymbol{X}_i$ implies $Z_i \perp\!\!\!\perp U_i$ conditional on $\boldsymbol{X}_i$.

This selection model setup is equivalent to the monotonicity condition, and is importing a well-known equivalence result from the IV literature to the CM setting. The main conceptual difference is not assuming Z_i is a valid instrument for identifying the D_i on Y_i effect among compliers; it is using the selection model representation to correct for selection bias. See Appendix A.6 for a validation of the general Vytlacil (2002) equivalence result in a CM setting, with conditioning covariates $\boldsymbol{X}_i$.

For notation purposes, suppose the vector of control variables $\boldsymbol{X}_i$ has at least two entries; denote $\boldsymbol{X}_i^{\text{IV}}$ as one entry in the vector, and $\boldsymbol{X}_i^{-}$ as the remaining. For each $z = 0, 1$ and each possible value of $\boldsymbol{x}_i^{\text{IV}}$ in the support of $\boldsymbol{X}_i^{\text{IV}}$, write the mediator propensity score as

$$\pi\left(z;\ \boldsymbol{x}^{\text{IV}}, \boldsymbol{x}^{-}\right) = \Pr\left(D_i = 1 \mid Z_i = z, \boldsymbol{X}_i^{\text{IV}} = \boldsymbol{x}^{\text{IV}}, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right).$$

Write $P_i = \pi(Z_i; \boldsymbol{X}_i^{-}, \boldsymbol{X}_i^{\text{IV}})$ for the observed propensity score, and $\mathcal{P}_z\left(\boldsymbol{X}_i^{-}\right) \coloneqq \text{Supp}\left(P_i \mid z, \boldsymbol{X}_i^{-}\right)$ for the IV-guided propensity score support for each treatment arm $z = 0, 1$.

Assumption MTE–2. Mediator take-up cost instrument: $\boldsymbol{X}_i^{\text{IV}}$ satisifes *relevance*, *exclusion*, *quasi-random assignment*, and *complier support*.

The first three are standard IV assumptions, and the last assumes for $z = 0, 1$ and each $(\boldsymbol{x}^{\text{IV}}, \boldsymbol{x}^{-})$ in the relevant support,

$$\left[\pi(0; \boldsymbol{x}^{\text{IV}}, \boldsymbol{x}^{-}),\ \pi(1; \boldsymbol{x}^{\text{IV}}, \boldsymbol{x}^{-})\right] \subseteq \text{Supp}\left(P_i \mid Z_i = z, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right).$$

Assumption MTE–2 is requiring at least one control variable guides selection-into-D_i — an Instrumental Variable (IV). Relevance provides variation in the mediator propensity score. Exclusion requires preceding healthcare access location to affect subsequent health and well-being only through healthcare take-up, after conditioning on $\boldsymbol{X}_i^{-}$ and Z_i. Quasi-random assignment requires this variation to be unrelated to latent mediator choices and mediator potential outcomes after conditioning on $\boldsymbol{X}_i^{-}$. Complier support requires variation in the mediator instrument, within each treatment arm and holding $\boldsymbol{X}_i^{-}$ fixed, to generate the propensity-score values associated with mediator take-up under both $Z_i = 0$ and $Z_i = 1$. It therefore permits the conditional outcome function within the $Z_i = z$ arm to be evaluated at both endpoints of the mediator-complier interval. Appendix A.7 gives these assumptions in full technical detail.

The exclusion and assignment conditions are substantive assumptions that must be defended using institutional knowledge. The relevance condition has empirical implications that can be evaluated using the mediator first stage. The most compelling example of a mediator

IV is using data on the cost of mediator take-up as a first-stage IV, if it varies between individuals for unrelated reasons and is strong in explaining mediator take-up. Complier support can be tested empirically among observed data, within each treatment value.

Assumption MTE–3. Common mediator-selection index.

Treatment Z_i and the mediator instrument $\boldsymbol{X}_i^{\mathrm{IV}}$ shift mediator take-up through the same scalar latent index U_i:

$$D_i(z, \boldsymbol{x}^{\mathrm{IV}}) = \mathbb{1}\left\{U_i \leq \pi(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{X}_i^-)\right\},$$

for $z = 0, 1$ and $\boldsymbol{x}^{\mathrm{IV}}$ on the $\boldsymbol{X}_i^{\mathrm{IV}}$ support, almost surely conditional on $\boldsymbol{X}_i^-$.

This assumption requires changes in Z_i and $\boldsymbol{X}_i^{\mathrm{IV}}$ to move the threshold on the same mediator-selection index. The mediator compliers induced by treatment therefore lie on the same latent margin traced by variation in the mediator instrument. Two-way monotonicity—monotonicity of mediator take-up in both treatment Z_i and the mediator instrument $\boldsymbol{X}_i^{\mathrm{IV}}$—is sufficient for representing their effects on mediator take-up through a common scalar resistance index U_i. In the OHIE, the assumption requires lottery selection and preceding healthcare access location to shift healthcare use along the same ordering of individuals by latent resistance to healthcare use.

For notation purposes, let K_z denote the number of distinct values taken by the mediator propensity score $\pi(z;\ \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^-)$, for each $z = 0, 1$.

Assumption MTE–CF. Restricted and separable Control Functions (CFs). The CFs belong to a $K_z - 1$ dimensional class over the complier distribution, and are separable from control variables, for $z = 0, 1$ and $p \in (0, 1)$

$$\mathbb{E}\left[U_i(z,0) \,\middle|\, p < U_i, \boldsymbol{X}_i^-\right] = \underbrace{\mathbb{E}\left[U_i(z,0) \,|\, p < U_i\right]}_{=:g_{z,0}(p)} + \boldsymbol{\phi}_{z,0}' \boldsymbol{X}_i^-,$$

$$\mathbb{E}\left[U_i(z,1) \,\middle|\, U_i \leq p, \boldsymbol{X}_i^-\right] = \underbrace{\mathbb{E}\left[U_i(z,1) \,|\, U_i \leq p\right]}_{=:g_{z,1}(p)} + \boldsymbol{\phi}_{z,1}' \boldsymbol{X}_i^-.$$

After observed heterogeneity is absorbed by $\mu_d(z; \boldsymbol{X}_i)$, then within each $Z_i = z$ remaining dependence between observed mediator take-up cost U_i and outcome gains $U_i(z,0), U_i(z,1)$ is common across $\boldsymbol{X}_i^-$ and lies in a $K_z - 1$ dimensional basis in the complier region, $\pi(0;\ \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^-)$ to $\pi(1;\ \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^-)$. Equivalently, the instrument identifies the CFs only up to the degrees of freedom supplied by variation in the instrument. For example, with only two relevant propensity-score values, this is a linear restriction on the CFs, corresponding to the restricted MTE logic of Brinch et al. (2017). This restricted support assumption is weaker than relying on identification at infinity, with testable implications — described further in Appendix A.8.

5.2 Re-identification of CM Effects

This section uses the above assumptions to identify the ADE and AIE in observed data, after using appropriate variation in mediator take-up supplied by the appropriate IV.

Write $m\big(p; z, \boldsymbol{X}_i^-\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = z, \boldsymbol{X}_i^-, P_i = p\right]$, the mean outcome given treatment, control values, and observed propensity score P_i. Assume it is finite for all values, and is absolutely continuous and its derivative exists almost everywhere.

Proposition 2. *If assumptions MTE–1, MTE–2 MTE–3, and MTE–CF hold, then the outcome mean conditional on the propensity score is identified in observed data*

$$m\big(p; z, \boldsymbol{X}_i^-\big) = \alpha + \gamma z + p\big(\beta + \delta z\big) + \phi^m(\boldsymbol{X}_i^-) + (1-p)g_{z,0}(p) + pg_{z,1}(p),$$

for each $p \in \mathcal{P}_z\left(\boldsymbol{X}_i^-\right)$ and $z = 0, 1$. alpha, β, γ, δ are the same parameters as in Section 4, and $g_{z,0}, g_{z,1}$ are the CFs defined above (with an assumed restricted shape).

Furthermore, the average Mediator MTE is point-wise identified on the propensity score support, as the difference between conditional means

$$m\big(p'; z, \boldsymbol{X}_i^-\big) - m\big(p\,; z, \boldsymbol{X}_i^-\big) = \int_p^{p'} \mathrm{MTE}(u; z, \boldsymbol{X}_i^-)\, du,$$

for any $p < p'$ such that $\big[p, p'\big] \subseteq \mathcal{P}_z(\boldsymbol{X}_i^-)$, and $z = 0, 1$. Proof: see Appendix **??**.

The mean difference of the mediator MTE is identified under the MTE Approach assumptions, thanks to instrumental variation in $\boldsymbol{X}_i^{\mathrm{IV}}$. The final step interpolates from the Mediator MTE to mediator compliers, and thus identifies the ADE and AIE.

Theorem MTE Approach. If assumptions MTE–1, MTE–2 MTE–3, and MTE–CF hold, then the ADE and AIE are identified as a function of the parameters in Proposition 2.

$$\begin{aligned} \mathrm{AIE}(z) &= \mathbb{E}\left[m\big(\pi(1; \boldsymbol{X}_i^-, \boldsymbol{X}_i^{\mathrm{IV}})\,; z, \boldsymbol{X}_i^-\big) - m\big(\pi(0; \boldsymbol{X}_i^-, \boldsymbol{X}_i^{\mathrm{IV}})\,; z, \boldsymbol{X}_i^-\big)\right] \\ \mathrm{ADE}(1-z) &= \mathrm{ATE} - \mathrm{AIE}(z), \qquad \text{for each } z = 0, 1. \end{aligned}$$

Proof: see Appendix A.9.

This theorem provides a solution to the identification problem for CM effects when facing selection. The MTE approach measures both selection bias and complier differences, and thus purges these persistent bias terms to identify CM effects.

6 MTE-based Estimation of CM Effects

The identification results suggest a direct three-step approach to estimating CM effects. First, estimate the mediator propensity score using variation in treatment, the mediator instrument, and the remaining control variables. Second, use the estimated propensity score to estimate the conditional-outcome function separately within each treatment arm. Third, evaluate this function at the two treatment-induced mediator propensities. Their difference estimates the AIE, while subtracting the estimated AIE from the ATE estimates the corresponding ADE.

For notational convenience, this section develops the estimator using the $Z_i = 0$ subsample for the $m(\cdot; 0, \boldsymbol{X}_i^-)$ estimation, which identifies AIE(0) and, consequently, ADE(1). The corresponding estimators of AIE(1) and ADE(0) follow by applying the same procedure, starting with the $Z_i = 1$ subsample.

6.1 Semi-parametric MTE

The first stage estimates the mediator propensity score,

$$\pi\big(z; \boldsymbol{X}_i^{\text{IV}}, \boldsymbol{X}_i^-\big) = \Pr\left(D_i = 1 \,\middle|\, Z_i = z, \boldsymbol{X}_i^{\text{IV}}, \boldsymbol{X}_i^-\right),$$

and constructs the generated regressor $\widehat{P}_i = \widehat{\pi}\big(Z_i; \boldsymbol{X}_i^{\text{IV}}, \boldsymbol{X}_i^-\big)$.

The propensity score can be estimated using a correctly specified binary-response model or an appropriate semi-parametric estimator. The identifying variation must come from $\boldsymbol{X}_i^{\text{IV}}$, conditional on treatment and the remaining controls, and must generate the propensity-score support required by Assumption MTE–2.

The second-stage now estimates the conditional-outcome function among the $Z_i = 0$ subsample,

$$m\big(p; 0, \boldsymbol{x}^-\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = 0, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

A convenient partially linear specification is

$$Y_i = \boldsymbol{\phi}_0' \boldsymbol{X}_i^- + s_0\big(\widehat{P}_i\big) + \varepsilon_{0,i}, \qquad \text{for } i \in \{j \,:\, Z_j = 0\},$$

where $s_0(.)$ is approximated using a polynomial, regression spline, or another finite-dimensional sieve basis. For a vector of known basis functions $\boldsymbol{b}_0(p)$, this can be written as

$$Y_i = \boldsymbol{\phi}_0' \boldsymbol{X}_i^- + \boldsymbol{b}_0\left(\widehat{P}_i\right)' \boldsymbol{\rho}_0 + \epsilon_{0,i}, \qquad \text{for } i : Z_i = 0.$$

The corresponding estimated conditional-outcome curve is constrcuted from the predictions, $\widehat{m}\big(p; 0, \boldsymbol{x}^-\big) = \widehat{\boldsymbol{\phi}}_0' \boldsymbol{x}^- + \boldsymbol{b}_0(p)' \widehat{\boldsymbol{\rho}}_0$.

The choice and dimension of $\boldsymbol{b}_0(p)$ must conform to the restricted-CF and rank conditions in Assumption MTE–CF. These conditions ensure that the finite IV-guided propensity-score support identifies the maintained approximation to the conditional-outcome curve over the mediator-complier interval. Appendix A.10.2 provides further details on the specification and estimation of this restricted outcome curve.[13]

Let $n_0 = \sum_{i=1}^{n}(1 - Z_i)$ denote the number of observations in the $Z_i = 0$ subsample. For each such observation, predict the $Z_i = 0$ conditional outcome at the mediator propensities induced by both treatment states:

$$\widehat{m}\left(\widehat{\pi}\left(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\right); 0, \boldsymbol{X}_i^{-}\right), \text{ and } \widehat{m}\left(\widehat{\pi}\left(0; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\right); 0, \boldsymbol{X}_i^{-}\right).$$

The difference between these predictions estimates the change in the $Z_i = 0$ outcome surface caused by moving the mediator propensity from its value under $Z_i = 0$ to its value under $Z_i = 1$. The sample-average estimator is

$$\widehat{\mathrm{AIE}}(0) = \frac{1}{n_0} \sum_{i:Z_i=0} \left[\widehat{m}\left(\widehat{\pi}\left(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\right); 0, \boldsymbol{X}_i^{-}\right) - \widehat{m}\left(\widehat{\pi}\left(0; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\right); 0, \boldsymbol{X}_i^{-}\right) \right].$$

Because Z_i is quasi-randomly assigned, the covariate distribution in the $Z_i = 0$ subsample consistently estimates the target population distribution. The preceding sample average therefore consistently standardises the estimated conditional indirect effects to the target population. A full-sample standardisation can alternatively be used as an efficiency refinement.

Evaluating the $Z_i = 0$ outcome curve at the propensity score induced by $Z_i = 1$ does not, by itself, constitute extrapolation. The relevant requirement is that both evaluation points lie in the domain identified by mediator-instrument variation within the $Z_i = 0$ arm. When an endpoint is an observed propensity-score support point, the corresponding conditional mean is directly identified. When an endpoint lies between supported values, its prediction is interpolation under the restricted-CF specification. The mediator-complier support condition rules out evaluation outside the identified domain.

The complementary direct effect is estimated using the decomposition $\mathrm{ATE} = \mathrm{AIE}(0) + \mathrm{ADE}(1)$. Thus, $\widehat{\mathrm{ADE}}(1) = \widehat{\mathrm{ATE}} - \widehat{\mathrm{AIE}}(0)$, where $\widehat{\mathrm{ATE}}$ is estimated from the quasi-random assignment of Z_i, using a differences in means (conditional on $\boldsymbol{X}_i^{-}$) by any standard estimator.

This endpoint estimator avoids numerical differentiation. The usual local-IV approach

[13] A parametric distributional assumption on the unobserved mediator-selection and outcome errors provides an alternative specification of $s_0(.)$. Joint normality, for example, produces the familiar inverse-Mills-ratio control function (Björklund & Moffitt 1987, Heckman 1979). The semi-parametric approach used here instead approximates $s_0(.)$ with a polynomial, spline, or sieve basis and therefore does not require a parametric distribution for the unobserved errors.

estimates the Mediator MTE from the derivative of a flexible conditional-outcome function with respect to the propensity score. Derivatives of estimated nonparametric functions can be unstable, particularly near sparse regions of the support. The present approach instead exploits the fact that the AIE is an integrated MTE: absolute continuity implies that its integral over the mediator-complier interval equals the difference between the conditional-outcome function evaluated at the two endpoints. Estimation therefore requires only fitted values of $\widehat{m}(.)$, rather than an estimated derivative.

Standard series and sieve results establish consistency of the fitted conditional-outcome curve under the maintained smoothness, approximation, support, and basis-growth conditions. Although the estimated function itself need not converge at a parametric rate, the sample-averaged endpoint difference is a smooth finite-dimensional functional and can be root-n consistent and asymptotically normal under additional regularity conditions. The effective sample size for estimating the $Z_i = 0$ outcome curve is n_0; if $\frac{n_0}{n}$ converges to a positive constant, normalisation by $\sqrt{n_0}$ and $\sqrt{n}$ differs only by a constant scale factor. Appendix A.10 states the relevant consistency and asymptotic-normality conditions.

Inference must account for the generated propensity score, which enters both as a regressor in the outcome-curve estimation and as the argument at which the fitted curve is evaluated. A practical approach is therefore to bootstrap the complete estimation procedure. Each bootstrap iteration re-estimates the mediator propensity score, the restricted $Z_i = 0$ outcome curve, both endpoint predictions, the AIE, and the ATE used to construct the ADE.

6.2 Simulation Evidence

The following simulation gives an example to show how these methods work in practice. Suppose data observed to the researcher $Z_i, D_i, Y_i, \boldsymbol{X}_i$ are drawn from the following data generating processes, for $i = 1, \ldots, N$, with $n = 5,000$ for this simulation.

$$Z_i \sim \text{Binom}\,(0.5)\,, \quad \boldsymbol{X}_i^{-} \sim N(4,1), \quad \boldsymbol{X}_i^{\text{IV}} \sim \text{Uniform}\,(-1,1)\,, \quad (U_i,(0,0),U_i(0,1),U_i(1,0),U_i(1,1),U_{C,i}) \sim I$$

$\boldsymbol{\Sigma}$ is the matrix of parameters which controls the level of confounding from unobserved costs and benefits.[14]

Each i chooses to take mediator D_i by a Roy model, with following mean definitions for

[14]The correlation and relative standard deviations for $U_i(.,0), U_i(.,1)$ affect how large selection bias in conventional CM estimates; correlation for these with unobserved costs $U_{C,i}$ does not particularly matter, though increased variance in unobserved costs makes estimates less precise for both OLS and MTE methods.

each $z, d = 0, 1$

$$D_i(z) = \mathbb{1}\left\{C_i \leq Y_i(z,1) - Y_i(z,0)\right\},$$
$$\mu_d\left(z; \boldsymbol{X}_i\right) = (z + d + zd) + \boldsymbol{X}_i^-, \quad \mu_C\left(z; \boldsymbol{X}_i\right) = 3z + \boldsymbol{X}_i^- - \boldsymbol{X}_i^{\text{IV}}.$$

Following Subsection 4.1, these data have the following first and second-stage equations:

$$D_i = \mathbb{1}\left\{U_{C,i} - \left(U_i(Z_i,1) - U_i(Z_i,0)\right) \leq 3z + \boldsymbol{X}_i^- - \boldsymbol{X}_i^{\text{IV}}\right\},$$
$$Y_i = Z_i + D_i + Z_i D_i + \boldsymbol{X}_i^- + \varepsilon_i.$$

Treatment Z_i has a causal effect on outcome Y_i, and it operates partially through mediator D_i. Outcome mean $\mu_{D_i}\left(Z_i; \boldsymbol{X}_i\right)$ contains an interaction term, $Z_i D_i$, so while Z_i, D_i have constant partial effects, the ATE depends on how many i choose to take the mediator and there is causal effect heterogeneity. ε_i collects the normally distributed residuals, as in Subsection 4.1.

After Z_i is assigned, i chooses to take mediator D_i by considering the costs and benefits — which vary based on Z_i, demographic controls $\boldsymbol{X}_i$, and the (non-degenerate) unobserved error terms $\Big(U_i(z,d)\Big)_{z,d=0,1}$. As a result, sequential quasi-random assignment does not hold; the mediator is not conditionally ignorable. Thus, a conventional approach to CM does not give an estimate for how much of the ATE goes through mediator D, but is contaminated by selection bias thanks to the unobserved error terms.

I simulate this data generating process 10,000 times, using

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.75 & 1 & 0.75 & 0 \\ 0.75 & 2.25 & 0.75 & 2.25 & 0 \\ 1 & 0.75 & 1 & 0.75 & 0 \\ 0.75 & 2.25 & 0.75 & 2.25 & 0 \\ 0 & 0 & 0 & 0 & 0.25 \end{pmatrix}.$$

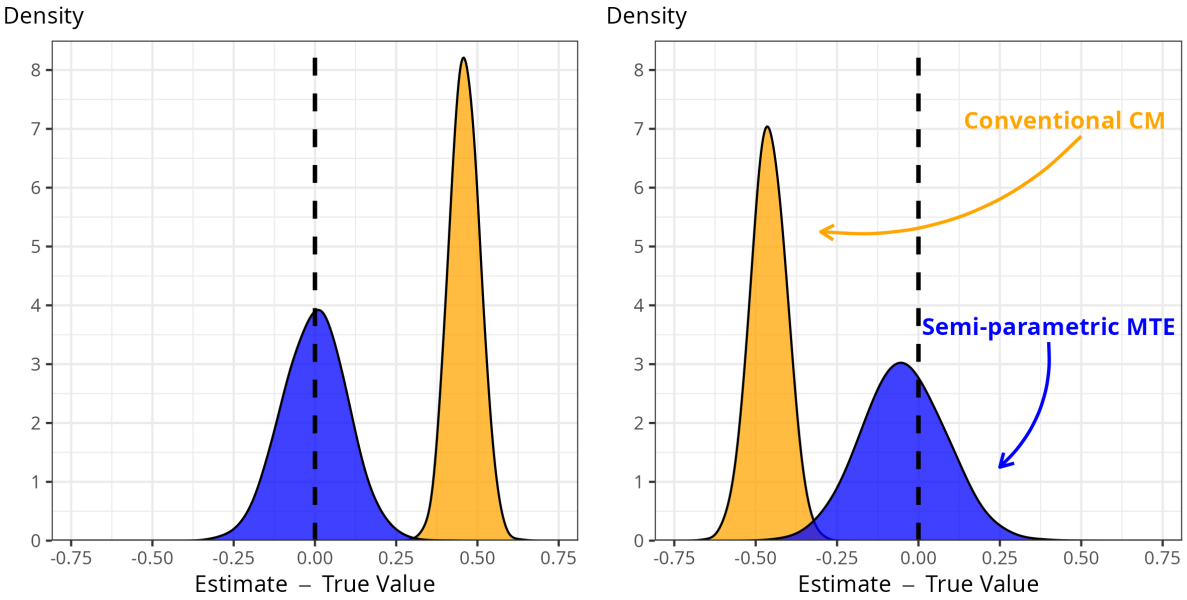
With these data simulated, I estimate CM effects with conventional CM methods (two-stage OLS) and the introduced MTE methods.[15] In this simulation $\Pr\left(D_i = 1\right) = 0.39$, and 45.8% of the sample are mediator compliers (for whom $D_i(0) = 0$ and $D_i(1) = 1$). This gives an ATE value of 2.13, AIE(0) 0.51 and ADE(1) 1.62, respectively.

Figure 4 shows how these estimates perform, with a semi-parametric MTE approach, relative to the true value. The OLS estimates' distribution do not overlap the true values for

[15] This choice of parameters has $U_i(0,d)$ following the same distribution as $U_i(1,d)$ (not differing across $Z_i = 0, 1$) for each $d = 0, 1$, and $\text{Var}\left(U_i(.,0)\right) = 1, \text{Var}\left(U_i(.,1)\right) = 2.25, \text{Corr}\big(U_i(.,0), U_i(.,1)\big) = 0.5$ so that unobserved errors meaningfully confound conventional CM methods, with notable heteroscedasticity. Unobserved costs are uncorrelated with $U_i(.,0), U_i(.,1)$ (although non-zero correlation would not meaningfully change the results), and $\text{Var}\left(U_{C,i}\right) = 0.25$ maintains uncertainty in unobserved costs.

Figure 4: The MTE Approach Addresses Persistent Bias in Conventional CM Estimates.

(a) $\widehat{\text{AIE}(0)} - \text{AIE}(0)$. **(b)** $\widehat{\text{ADE}(1)} - \text{ADE}(1)$.



Note: These figures show the empirical density of point estimates minus the true average effect, for 10,000 different datasets generated from a Roy model with normally distributed error terms (with both correlated error terms and heteroscedasticity, further described in Subsection 6.2). The black dashed line is the true value; orange is the distribution of conventional CM estimates from two-stage OLS (Imai et al. 2010), and blue estimates with a two-stage Heckman selection adjustment.

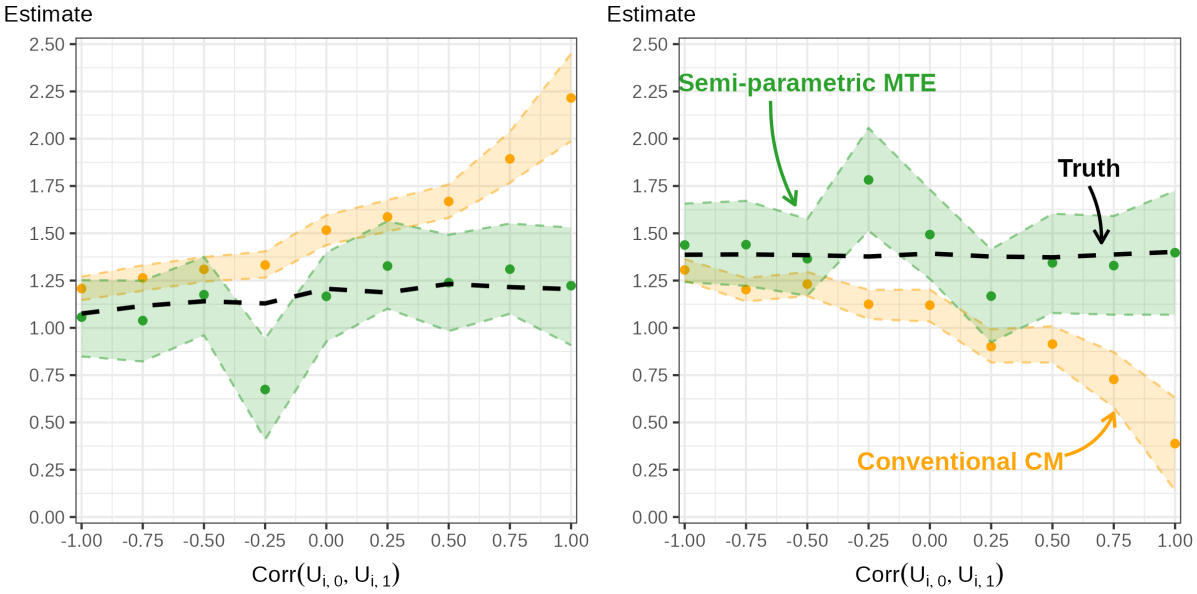
any standard level of significance; the distance between the OLS estimates and the true values are the underlying bias terms derived in Theorem 1. The semi-parametric MTE approach perfectly reproduces the true values, as the probit first-stage correctly models the normally distributed error terms.

The error terms determine the bias in OLS estimates of the ADE and AIE, so the bias varies for different values of the error-term parameters $\text{Corr}\big(U_i(z,0), U_i(.,1)\big) \in [-1,1]$ and $U_i(Z_i,1) \geq 0$. The true AIE values vary, because $D_i(Z_i)$ compliers have higher average values of $U_i(Z_i,1) - U_i(Z_i,0)$ as $\text{Corr}\big(U_i(Z_i,0), U_i(Z_i,1)\big)$ increases. Figure 5 shows MTE-based estimates against estimates calculated by standard OLS, showing 95% confidence intervals calculated from 1,000 bootstraps. The point estimates of the MTE-based approach do not exactly equal the true values, as they are estimates from one simulation. The MTE approach improves on OLS estimates by correcting for bias, with confidence regions overlapping the true values.

This correction did not come for free: the standard errors are significantly greater in an MTE approach than conventional CM estimates (based on two-stage OLS). In this manner, this simulation shows the pros and cons of using the MTE approach to estimating CM effects

Figure 5: MTE-Based Estimates Work with Different Error Term Parameters.

(a) AIE(0). **(b)** ADE(1).



Note: These figures show the OLS and MTE-based point estimates of the AIE(0) and ADE(1), for $n = 5,000$ sample size, varying $\text{Corr}\big(U_i(z,0), U_i(z,1)\big)$ values with $\text{Var}\left(U_i(z,d)\right) = 1$ fixed for all $z, d = 0, 1$. The black dashed line is the true value, coloured points are points estimates for the respective data generated, and shaded regions are the 95% confidence intervals from 1,000 bootstraps each. Orange represents OLS estimates, blue the semi-parametric MTE approach.

in practice.

7 CM in the Oregon Health Insurance Experiment

In the Oregon Health Insurance Experiment, winning the wait-list lottery significantly improved subjective health and well-being among participants. This study investigates the mechanisms behind these benefits, quantifying the extent to which improvements are mediated through increased healthcare usage.

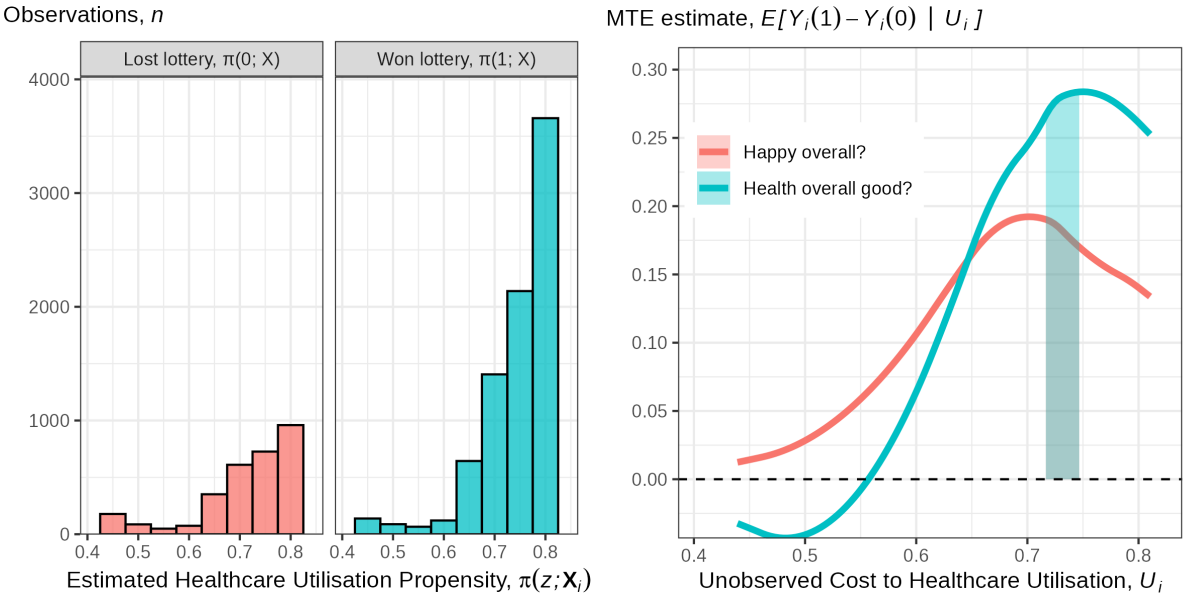
To address concerns for unobserved selection-into-healthcare, I use the respondents' regular healthcare provider before the wait-list lottery as an IV for healthcare usage. Approximately 73.2% reported visiting a healthcare provider within the past year, but rates vary notably depending on their provider type: those who reported attended hospital emergency rooms (A&E) and urgent care clinics before the wait-list lottery reported significantly lower healthcare visitation rates after the lottery (8.4 and 11 percentage points lower, respectively) than the 40% who attended private clinics.[16] The IV validity arises from differential costs

[16]The combined F statistic for the categorical variable for healthcare usual location (before the wait-list

faced by individuals based on their usual care provider. Private clinics generally charge through health insurance and are more expensive without coverage, while A&E and urgent care often provide costly services but rarely follow up on unpaid bills, effectively creating variation in healthcare attendance costs. Additionally, individuals' choice of provider likely depend on neighbourhood-based access.

Figure 6: Two-Stage Estimates in the Oregon Health Insurance Experiment.

(a) First-stage Propensity Score Estimates. **(b)** MTE Estimates.



Note: Panel A shows the estimates of the first-stage propensity to use healthcare in the year following the health insurance experiment. It uses a logit specification, controlling for household size and whether losing the lottery $\pi(0; \boldsymbol{X}_i)$, or winning the lottery $\pi(1; \boldsymbol{X}_i)$. Average utilisaition after having lost the lottery is roughly $\mathbb{E}[D_i \mid Z_i = 0] = \mathbb{E}[\pi(0; \boldsymbol{X}_i)] = 0.715$, and after winning $\mathbb{E}[D_i \mid Z_i = 1] = \mathbb{E}[\pi(1; \boldsymbol{X}_i)] = 0.750$. Panel B shows an estimated point-wise Mediator MTE in the lottery loser $Z_i = 0$ subsample, using local IV for the estimation procedure. The shaded region refers to the AIE(0), where the average causal effect is calculated by integrating across the region of mediator compliers, with $\pi(0; \boldsymbol{X}_i) \leq U_i < \pi(1; \boldsymbol{X}_i)$ — i.e., between 0.715 and 0.750 at the average covariate values.

Initial results with unadjusted CM estimates suggest almost no mediating role for healthcare usage; the unadjusted estimates of the AIE are close to zero for both outcomes, contradicting intuitive suggestive evidence. These estimates control for diagnosis of serious health conditions (before the wait-list lottery), such as kidney disease or diabetes. However, this approach would not be considering possible unobserved confounding coming from underlying health conditions. My MTE-based approach attempts to model this unobserved confounding, and applying them to these data reveals a much larger, positive AIE, restoring the mediating mechanism of healthcare usage in line with suggestive intuition. This is because a

lottery) on healthcare usage (following the lottery) is 38.4.

correlational estimate of health and well-being gains to healthcare visits are practically zero, while the IV estimates restore positive gains and the MTE-based methods pick this up with a larger AIE estimate. These numbers are reported in Table 1, where panel A shows the CM effects with the binary outcome of subjective health, and panel B the binary outcome of subjective well-being.

Table 1: CM Effect Estimates for Wait-list Lottery Effects on Health and Well-being.

	First-stage	ATE	ADE(1)	AIE(0)	AIE(0) / ATE
	(1)	(2)	(3)	(4)	(5)
Panel A: Health overall good?					
Unadjusted	4.700	4.500	4.700	-0.190	-0.043
	(0.880)	(0.910)	(0.910)	(0.063)	(0.018)
Parametric MTE	4.70	5.00	2.50	1.80	0.36
	(0.88)	(0.95)	(1.20)	(0.55)	(0.14)
Semi-parametric MTE	4.70	5.00	2.00	3.10	0.61
	(0.84)	(0.97)	(1.30)	(0.87)	(0.22)
Panel B: Happy overall?					
Unadjusted	4.7000	7.1000	7.0000	0.0670	0.0094
	(0.8600)	(0.9600)	(0.9600)	(0.0520)	(0.0077)
Parametric MTE	4.70	7.50	5.00	1.90	0.25
	(0.870)	(0.990)	(1.100)	(0.510)	(0.075)
Semi-parametric MTE	4.70	7.50	5.00	2.50	0.34
	(0.84)	(0.93)	(1.20)	(0.74)	(0.11)

Note: This table shows the point estimates (and SEs in brackets) of applying the proposed CM methods to replication data from the Oregon Health Insurance Experiment (Finkelstein & Baicker 2014). The first-stage column refers to the average effect of winning the wait-list lottery on healthcare usage (mediator first-stage), ATE average effect on surveyed health and well-being, AIE and ADE to respective CM effects through and absent healthcare usage. SEs were calculated with 1,000 bootstrap replications. The numbers are pp increases in the binary outcome, so an estimate of 4.7 in row 1 column 1 means an increase in 4.7 pp of using healthcare in the last 12 months after winning the wait-list lottery.

This reversal in conclusions highlights the importance of correcting for negative selection into healthcare usage. A conventional approach to CM fails to account for the fact that individuals with poorer underlying health tend to visit healthcare providers more frequently, generating negative selection bias that obscures the true positive AIE, and clouds inference on the mediator mechanism. By explicitly adjusting for this bias using the MTE approach, I isolate a credible positive indirect effect of healthcare usage on subjective health and well-being.

These findings offer credible evidence that improved healthcare access yields meaningful

subjective health and well-being benefits, despite previous research emphasising negligible effects on objective health measures such as blood pressure (Baicker, Taubman, Allen, Bernstein, Gruber, Newhouse, Schneider, Wright, Zaslavsky & Finkelstein 2013). Subjective measures likely reflect broader psychological and financial relief associated with reduced healthcare-related anxiety and diminished risk of catastrophic medical debt, thus producing more noticeable short-term subjective improvements.

Nevertheless, this analysis is subject to notable limitations. The IV is not ideal, and potentially more important mediators (such as explicit health insurance status) would require additional IVs beyond the wait-list lottery itself, presenting a challenging identification issue. Furthermore, the 95% confidence intervals for both ADE and AIE estimates (based on bootstrapped SEs) remain large, though statistically significant and excluding zero. This uncertainty underscores common challenges in applied CM analyses, where statistical precision can be limited by data constraints.

8 Summary and Concluding Remarks

This paper has studied a selection-on-observables approach to CM in a natural experiment setting. I have shown the pitfalls of using the most popular methods for estimating direct and indirect effects without a clear case for the mediator being quasi-randomly assigned. Using the Roy model as a benchmark, a mediator is unlikely to be quasi-randomly assigned in natural experiment settings, and the bias terms likely crowd out inference regarding CM effects.

This paper has also contributed to the growing CM literature in economics, connecting to MTE methods and developing a compelling way of estimating direct and indirect effects in a natural experiment setting. It has also recognised limitations in the common practice of suggestive evidence for mechanisms, and given credible CM estimates in a famous natural experiment setting well-known in the economics field. Further research could build on the approaches presented by suggesting efficiency improvements, adjustments for common statistical irregularities (say, cluster dependence), or integrating the MTE approach to the growing double robustness literature on CM (Farbmacher, Huber, Laffers, Langen & Spindler 2022, Bia, Huber & Laffers 2024).

These findings do not provide a blanket endorsement for applied researchers to use CM methods. There are strong structural assumptions for adjusting identifying CM effects despite unobserved selection-into-mediator, and inference requires an IV for mediator take-up. If these assumptions do not hold true, then selection-adjusted estimates of CM effects will also be biased, and will not improve on an unadjusted conventional approach.

Yet, there are likely settings in which the structural assumptions are credible. Mediator monotonicity aligns well with economic theory in many cases, and it is plausible for researchers to study big data settings with external variation in mediator take-up costs. In these cases, this paper opens the door to identifying mechanisms behind treatment effects in natural experiment settings.

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A Supplementary Appendix

This section is for supplementary information, and validation of presented propositions and theorems. It is not meant for publication.

Any comments or suggestions may be sent to me at seh325@cornell.edu, or raised as an issue on the Github project, https://github.com/shoganhennessy/mediation-natural-experiment.

A.1 Identification in Causal Mediation

Imai et al. (2010, Theorem 1) states that the ADE and AIE are identified under sequential quasi-random assignment. Write $F_{\boldsymbol{X}_i}$ for the cumulative density function for control variables $\boldsymbol{X}_i$. For each $z = 0, 1$:

$$\mathbb{E}\left[Y_i(1, D_i(z)) - Y_i(0, D_i(z))\right] = \int \int \Big(\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i, \boldsymbol{X}_i\right] \Big) dF_{D_i \,|\, Z_i = z, \boldsymbol{X}_i} dF_{\boldsymbol{X}_i},$$
$$\mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0))\right] = \int \int \mathbb{E}\left[Y_i \,|\, Z_i = z, D_i, \boldsymbol{X}_i\right] \Big(dF_{D_i \,|\, Z_i = 1, \boldsymbol{X}_i} - dF_{D_i \,|\, Z_i = 0, \boldsymbol{X}_i} \Big) dF_{\boldsymbol{X}_i}.$$

In the setting with $D_i = 0, 1$ binary considered in this paper, these are written as sums instead. For each $z = 0, 1$,

$$\begin{aligned} \text{ADE}(z) &= \mathbb{E}\left[Y_i(1, D_i(z)) - Y_i(0, D_i(z))\right] \\ &= \int \sum_{d \in \{0,1\}} \Big(\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i = d, \boldsymbol{X}_i = \boldsymbol{x}\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = d, \boldsymbol{X}_i\right] \Big) \\ &\qquad \times \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \, dF_{\boldsymbol{X}_i}. \end{aligned}$$

Likewise, for the average natural indirect effect, for each $z = 0, 1$

$$\begin{aligned} \text{AIE}(z) &= \mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0))\right] \\ &= \int \sum_{d \in \{0,1\}} \mathbb{E}\left[Y_i \,|\, Z_i = z, D_i = d, \boldsymbol{X}_i\right] \\ &\qquad \times \Big(\Pr\left(D_i = d \,|\, Z_i = 1, \boldsymbol{X}_i\right) - \Pr\left(D_i = d \,|\, Z_i = 0, \boldsymbol{X}_i = \boldsymbol{x}\right) \Big) \, dF_{\boldsymbol{X}_i}. \end{aligned}$$

A.2 Bias in Causal Mediation (CM) Estimands

Suppose that Z_i is ignorable conditional on $\boldsymbol{X}_i$, but D_i is not.

A.2.1 Bias in the Average Direct Effect (ADE)

Consider the conventional estimand for ADE(z), for each $z = 0, 1$.

$$\begin{aligned}&\mathbb{E}_{D_i \mid Z_i=z, \boldsymbol{X}_i}\left[\mathbb{E}\left[Y_i \mid Z_i = 1, D_i, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i, \boldsymbol{X}_i\right]\right] \\ &= \sum_{d \in \{0,1\}} \Pr\left(D_i = d \mid Z_i = z, \boldsymbol{X}_i\right)\Big(\mathbb{E}\left[Y_i \mid Z_i = 1, D_i = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i = d, \boldsymbol{X}_i\right]\Big).\end{aligned}$$

Conditional on $\boldsymbol{X}_i$, it averages the outcome contrast between $Z_i = 1$ and $Z_i = 0$ at each fixed mediator value d over the mediator distribution under treatment state z.

Fix $d \in \{0, 1\}$. Treatment quasi-random assignment and consistency give,

$$\begin{aligned}\mathbb{E}\left[Y_i \mid Z_i = 1, D_i = d, \boldsymbol{X}_i\right] &= \mathbb{E}\left[Y_i(1, d) \mid D_i(1) = d, \boldsymbol{X}_i\right], \\ \mathbb{E}\left[Y_i \mid Z_i = 0, D_i = d, \boldsymbol{X}_i\right] &= \mathbb{E}\left[Y_i(0, d) \mid D_i(0) = d, \boldsymbol{X}_i\right].\end{aligned}$$

Therefore,

$$\begin{aligned}&\mathbb{E}\left[Y_i \mid Z_i = 1, D_i = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i = d, \boldsymbol{X}_i\right] \\ &= \mathbb{E}\left[Y_i(1, d) \mid D_i(1) = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(0, d) \mid D_i(0) = d, \boldsymbol{X}_i\right].\end{aligned}$$

Add and subtract $\mathbb{E}\left[Y_i(0, d) \mid D_i(1) = d, \boldsymbol{X}_i\right]$, to give

$$\begin{aligned}&\mathbb{E}\left[Y_i \mid Z_i = 1, D_i = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i = d, \boldsymbol{X}_i\right] \\ &= \mathbb{E}\left[Y_i(1, d) - Y_i(0, d) \mid D_i(1) = d, \boldsymbol{X}_i\right] \\ &\quad + \mathbb{E}\left[Y_i(0, d) \mid D_i(1) = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(0, d) \mid D_i(0) = d, \boldsymbol{X}_i\right].\end{aligned}$$

The first term is the direct effect with the mediator held at d among individuals for whom $D_i(1) = d$. The second term is selection bias arising from differences in untreated potential outcomes between individuals for whom $D_i(1) = d$ and individuals for whom $D_i(0) = d$.

To relate the first term to ADE(z), note that

$$\Pr\left(D_i = d \mid Z_i = z, \boldsymbol{X}_i\right) = \Pr\left(D_i(z) = d \mid \boldsymbol{X}_i\right),$$

Unfix d, to note the target conditional ADE as a weighted average of fixed-mediator direct

effects local to the individuals with $D_i(z) = d$:

$$\begin{aligned}
&\mathbb{E}\left[Y_i(1, D_i(z)) - Y_i(0, D_i(z)) \,|\, \boldsymbol{X}_i\right] \\
&= \sum_{d \in \{0,1\}} \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(1,d) - Y_i(0,d) \,|\, Z_i = z, D_i = d, \boldsymbol{X}_i\right] \\
&= \sum_{d \in \{0,1\}} \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(1,d) - Y_i(0,d) \,|\, D_i(z) = d, \boldsymbol{X}_i\right]
\end{aligned}$$

where the second equality follows from treatment quasi-random assignment.

Substituting the fixed-d decomposition into the conventional CM estimand gives

$$\begin{aligned}
&\mathbb{E}_{D_i \,|\, Z_i = z, \boldsymbol{X}_i}\left[\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i, \boldsymbol{X}_i\right]\right] \\
&= \sum_{d \in \{0,1\}} \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \Big(\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = d, \boldsymbol{X}_i\right]\Big) \\
&= \sum_{d \in \{0,1\}} \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(1,d) - Y_i(0,d) \,|\, D_i(1) = d, \boldsymbol{X}_i\right] \\
&\quad + \sum_{d \in \{0,1\}} \Pr\left(D_i = d \,|\, Z_i = z, \boldsymbol{X}_i\right) \Big(\mathbb{E}\left[Y_i(0,d) \,|\, D_i(1) = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(0,d) \,|\, D_i(0) = d, \boldsymbol{X}_i\right]\Big).
\end{aligned}$$

Separate the group difference from selection by add and subtract the conditional ADE, and average over $\boldsymbol{X}_i$ to give the final terms

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{X}_i}\left[\mathbb{E}_{D_i \,|\, Z_i = z, \boldsymbol{X}_i}\left[\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i, \boldsymbol{X}_i\right]\right]\right] \\
&= \mathrm{ADE}(z) \\
&\quad + \underbrace{\mathbb{E}_{\boldsymbol{X}_i}\left[\mathbb{E}_{D_i = d \,|\, Z_i = z, \boldsymbol{X}_i}\left[\mathbb{E}\left[Y_i(0,d) \,|\, D_i(1) = d,\ \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(0,d) \,|\, D_i(0) = d,\ \boldsymbol{X}_i\right]\right]\right]}_{\text{Selection bias}} \\
&\quad + \underbrace{\mathbb{E}_{\boldsymbol{X}_i}\left[\mathbb{E}_{D_i = d \,|\, Z_i = z, \boldsymbol{X}_i}\left[\mathbb{E}\left[Y_i(1,d) - Y_i(0,d) \,|\, D_i(1) = d, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(1,d) - Y_i(0,d) \,|\, D_i(z) = d, \boldsymbol{X}_i\right]\right]\right]}_{\text{Group-difference bias.}}
\end{aligned}$$

Note that in the setting $z = 1$, the ADE(1) group-difference term equal to zero, group-difference term equal to zero, because both conditional direct effects are evaluated among individuals for whom $D_i(1) = d$.

A.2.2 Bias in the Average Indirect Effect (AIE)

To show that the conventional approach to mediation gives an estimate for the AIE with selection and group difference-bias, start with the definition of the AIE — the indirect effect among compliers times the size of the complier group. This sequence relies on assuming no mediator defiers; the following appendix subsection adds another bias term in the case that

mediator defiers exist.

Fix $z \in \{0, 1\}$. This proof starts with the conditional AIE, conditional on a specific value of $\boldsymbol{X}_i$.

$$\begin{aligned}
&\mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0)) \mid \boldsymbol{X}_i\right] \\
&= \Pr\left(D_i(0) = 0, D_i(1) = 1 \mid \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]
\end{aligned}$$

When D_i is not ignorable, the bias comes from attempting to identify the second term, $\mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]$, the indirect effect among mediator compliers.

Again, note the mean outcomes in terms of average potential outcomes,

$$\begin{aligned}
\mathbb{E}\left[Y_i \mid Z_i = z, D_i = 1, \boldsymbol{X}_i\right] &= \mathbb{E}\left[Y_i(z, 1) \mid D_i(z) = 1, \boldsymbol{X}_i\right], \\
\mathbb{E}\left[Y_i \mid Z_i = z, D_i = 0, \boldsymbol{X}_i\right] &= \mathbb{E}\left[Y_i(z, 0) \mid D_i(z) = 0, \boldsymbol{X}_i\right].
\end{aligned}$$

Compose the selection bias term, as follows.

$$\begin{aligned}
&\mathbb{E}\left[Y_i \mid Z_i = z, D_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = z, D_i = 0, \boldsymbol{X}_i\right] \\
&= \mathbb{E}\left[Y_i(z, 1) \mid D_i(z) = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(z, 0) \mid D_i(z) = 0, \boldsymbol{X}_i\right] \\
&= \mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] + \mathbb{E}\left[Y_i(z, 0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(z, 0) \mid D_i(z) = 0, \boldsymbol{X}_i\right]
\end{aligned}$$

The right-hand side is the sum of the mediator effect among individuals with $D_i(z) = 1$ and a selection-bias term capturing differences in baseline potential outcomes between individuals with $D_i(z) = 1$ and $D_i(z) = 0$

The AIE is the indirect effect among compliers times the size of the complier group, so we need to compensate for the difference between the med $D_i(z) = 1$ and mediator-complier group $D_i(0) = 0, D_i(1) = 1$.

$$\begin{aligned}
&\mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] \\
&= \mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right] \\
&\quad + \mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(z, 1) - Y_i(z, 0) \mid D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right].
\end{aligned}$$

The first term is the mediator effect among mediator compliers, while the remaining difference is the group-difference bias from using the mediator effect among individuals with $D_i(z) = 1$ in its place.

Collect everything together, as follows.

$$
\begin{aligned}
&\mathbb{E}\left[Y_i \mid Z_i = z, D_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i = z, D_i = 0, \boldsymbol{X}_i\right] \\
&= \underbrace{\mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \mid D_i(1) = 1, D_i(0) = 0, \boldsymbol{X}_i\right]}_{\text{Effect of } D_i \text{ among compliers, conditional on } \boldsymbol{X}_i} \\
&+ \underbrace{\mathbb{E}\left[Y_i(z,0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(z,0) \mid D_i(z) = 0, \boldsymbol{X}_i\right]}_{\text{Selection bias}} \\
&+ \underbrace{\mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \mid D_i(z) = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \mid D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]}_{\text{group difference-bias}}
\end{aligned}
$$

The proof is finally achieved by multiplying by the complier score, $\Pr\left(D_i(0) = 0, D_i(1) = 1 \mid \boldsymbol{X}_i\right)$ $= \mathbb{E}\left[D_i \mid Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \mid Z_i = 0, \boldsymbol{X}_i\right]$, then averaging across $\boldsymbol{X}_i$.

A.3 The AIE Without Mediator Monotonicity

The main text maintains mediator monotonicity,

$$\Pr\left(D_i(0) \leq D_i(1)\right) = 1.$$

This assumption rules out mediator defiers and allows the mediator first-stage to be interpreted as the mediator complier share. This section relaxes mediator monotonicity and shows how the AIE decomposition changes when mediator defiers are present.

For each value of $\boldsymbol{X}_i$, define the mediator complier and defier shares as

$$\pi_C(\boldsymbol{X}_i) := \Pr\big(D_i(0) = 0, D_i(1) = 1 \mid \boldsymbol{X}_i\big),$$

and

$$\pi_D(\boldsymbol{X}_i) := \Pr\big(D_i(0) = 1, D_i(1) = 0 \mid \boldsymbol{X}_i\big).$$

The mediator first-stage is then

$$
\begin{aligned}
&\mathbb{E}\left[D_i \mid Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \mid Z_i = 0, \boldsymbol{X}_i\right] \\
&\quad = \mathbb{E}\left[D_i(1) - D_i(0) \mid \boldsymbol{X}_i\right] \\
&\quad = \pi_C(\boldsymbol{X}_i) - \pi_D(\boldsymbol{X}_i). \qquad (7)
\end{aligned}
$$

Thus, absent monotonicity, the mediator first-stage identifies the net complier share rather than the mediator complier share.

Fix $z \in \{0, 1\}$. Because $D_i(0)$ and $D_i(1)$ are binary,

$$Y_i\big(z, D_i(1)\big) - Y_i\big(z, D_i(0)\big) = \big[D_i(1) - D_i(0)\big]\big[Y_i(z,1) - Y_i(z,0)\big].$$

Always-takers and never-takers make no contribution to this expression because their mediator status does not respond to Z_i. Compliers contribute positively, while defiers contribute negatively.

Define the average effects of the mediator among compliers and defiers, respectively, as

$$\tau_C(z; \boldsymbol{X}_i) \coloneqq \mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \,|\, D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]$$

$$\tau_D(z; \boldsymbol{X}_i) \coloneqq \mathbb{E}\left[Y_i(z,1) - Y_i(z,0) \,|\, D_i(0) = 1, D_i(1) = 0, \boldsymbol{X}_i\right].$$

The conditional AIE can therefore be written as

$$\mathbb{E}\left[Y_i\big(z, D_i(1)\big) - Y_i\big(z, D_i(0)\big) \,\middle|\, \boldsymbol{X}_i\right] = \pi_C(\boldsymbol{X}_i)\tau_C(z; \boldsymbol{X}_i) - \pi_D(\boldsymbol{X}_i)\tau_D(z; \boldsymbol{X}_i).$$

Applying the expectation across $\boldsymbol{X}_i$ gives

$$\text{AIE}(z) = \mathbb{E}\left[\pi_C(\boldsymbol{X}_i)\tau_C(z; \boldsymbol{X}_i) - \pi_D(\boldsymbol{X}_i)\tau_D(z; \boldsymbol{X}_i)\right]. \tag{8}$$

Using Equation (7), this can alternatively be written as

$$\begin{aligned}
&\mathbb{E}\left[Y_i\big(z, D_i(1)\big) - Y_i\big(z, D_i(0)\big) \,\middle|\, \boldsymbol{X}_i\right] \\
&\quad = \Big(\mathbb{E}\left[D_i \,|\, Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \,|\, Z_i = 0, \boldsymbol{X}_i\right]\Big)\tau_C(z; \boldsymbol{X}_i) \\
&\qquad + \pi_D(\boldsymbol{X}_i)\left[\tau_C(z; \boldsymbol{X}_i) - \tau_D(z; \boldsymbol{X}_i)\right].
\end{aligned} \tag{9}$$

The first term is the net mediator first-stage multiplied by the average mediator effect among compliers. The second term adjusts for differences between the average mediator effects of compliers and defiers.

Under mediator monotonicity, $\pi_D(\boldsymbol{X}_i) = 0$, and equation (9) reduces to the complier-based decomposition used in the main text. The additional term also vanishes if compliers and defiers have the same average mediator effect, although this is a treatment-effect homogeneity restriction rather than monotonicity. Without either restriction, the mediator first-stage cannot be interpreted as the mediator complier share, and the AIE cannot be written as the mediator first-stage multiplied only by the average mediator effect among compliers.

The presence of mediator defiers also changes the selection-bias decomposition for the conventional CM estimand. Absent mediator monotonicity, the conventional CM estimand

for the AIE can be written as

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{X}_i}\Big[\Big(\mathbb{E}\left[D_i \,|\, Z_i=1,\boldsymbol{X}_i\right]-\mathbb{E}\left[D_i \,|\, Z_i=0,\boldsymbol{X}_i\right]\Big)\Big(\mathbb{E}\left[Y_i \,|\, Z_i=z,D_i=1,\boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i \,|\, Z_i=z,D_i=0,\boldsymbol{X}_i\right]\Big)\Big] \\
&= \mathrm{AIE}(z) \\
&\quad + \underbrace{\mathbb{E}_{\boldsymbol{X}_i}\Big[\Big(\pi_C(\boldsymbol{X}_i)-\pi_D(\boldsymbol{X}_i)\Big)\Big(\mathbb{E}\left[Y_i(z,0) \,|\, D_i(z)=1,\boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i(z,0) \,|\, D_i(z)=0,\boldsymbol{X}_i\right]\Big)\Big]}_{\text{Selection bias}} \\
&\quad + \underbrace{\mathbb{E}_{\boldsymbol{X}_i}\Big[\Big(\pi_C(\boldsymbol{X}_i)-\pi_D(\boldsymbol{X}_i)\Big)\Big(\mathbb{E}\left[Y_i(z,1)-Y_i(z,0) \,|\, D_i(z)=1,\boldsymbol{X}_i\right]-\tau_C(z;\boldsymbol{X}_i)\Big)\Big]}_{\text{Group-difference bias}} \\
&\quad + \underbrace{\mathbb{E}_{\boldsymbol{X}_i}\Big[\pi_D(\boldsymbol{X}_i)\Big(\tau_D(z;\boldsymbol{X}_i)-\tau_C(z;\boldsymbol{X}_i)\Big)\Big]}_{\text{Defier compensation}}.
\end{aligned}
$$

The second term is selection bias: differences in baseline potential outcomes between those who take and refuse the mediator. The third term is group-difference bias: the average mediator effect among those observed taking the mediator may differ from the average mediator effect among mediator compliers. Both terms are weighted by the net mediator first-stage, $\pi_C(\boldsymbol{X}_i)-\pi_D(\boldsymbol{X}_i)$, rather than the mediator complier share alone. The final term is the additional defier-compensation term. It arises because the true AIE contains the negatively weighted mediator effect among defiers, while the conventional CM estimand weights its second-stage mediator contrast by the net mediator first-stage.

A.4 A Regression Framework for Direct and Indirect Effects

The first-stage regression of $Z \to Y$ has unbiased estimates, since $Z_i \perp\!\!\!\perp D_i(.)\big|\boldsymbol{X}_i$. Put $\pi(z;\boldsymbol{X})=\mathbb{E}\left[D_i(z) \,|\, \boldsymbol{X}\right]$, and $\eta_{z,i}=D_i(z)-\pi(z;\boldsymbol{X})$ the first-stage error terms.

$$
\begin{aligned}
D_i &= Z_iD_i(1)+(1-Z_i)D_i(0) \\
&= D_i(0)+Z_i\left[D_i(1)-D_i(0)\right] \\
&= \underbrace{\pi(0;\boldsymbol{X}_i)}_{\text{Intercept, }:=\theta+\zeta(\boldsymbol{X}_i)} + \underbrace{Z_i\big(\pi(1;\boldsymbol{X}_i)-\pi(0;\boldsymbol{X}_i)\big)}_{\text{Regressor, }:=\overline{\pi}Z_i} + \underbrace{(1-Z_i)\eta_{0,i}+Z_i\eta_{1,i}}_{\text{Errors, }:=\eta_i}
\end{aligned}
$$

$$
\implies \mathbb{E}\left[D_i \,|\, Z_i,\boldsymbol{X}_i\right]=\theta+\overline{\pi}Z_i+\zeta(\boldsymbol{X}_i).
$$

Since the quasi-random assignment assumption gives $\mathbb{E}\left[Z_i\eta_{z,i} \,|\, \boldsymbol{X}_i\right]=\mathbb{E}\left[Z_i \,|\, \boldsymbol{X}_i\right]\mathbb{E}\left[\eta_{z,i} \,|\, \boldsymbol{X}_i\right]=0$, for each $z=0,1$.

The same cannot be said of the regression that estimates direct and indirect effects, without further assumptions. Write the mean potential outcome as $\mu_d(z;\boldsymbol{X})=\mathbb{E}\left[Y_i(z,d) \,|\, \boldsymbol{X}\right]$, with

residual $U_i(z,d) = Y_i(z,d) - \mu_d(z;\boldsymbol{X})$ for each $z, d = 0, 1$.

$$\begin{aligned}
Y_i &= Z_i Y_i(1, D_i(1)) + (1 - Z_i) Y_i(0, D_i(0)) \\
&= Z_i D_i Y_i(1,1) \\
&\quad + (1 - Z_i) D_i Y_i(0,1) \\
&\quad + Z_i (1 - D_i) Y_i(1,0) \\
&\quad + (1 - Z_i)(1 - D_i) Y_i(0,0) \\
&= Y_i(0,0) \\
&\quad + Z_i \left[Y_i(1,0) - Y_i(0,0) \right] \\
&\quad + D_i \left[Y_i(0,1) - Y_i(0,0) \right] \\
&\quad + Z_i D_i \left[Y_i(1,1) - Y_i(1,0) - (Y_i(0,1) - Y_i(0,0)) \right]
\end{aligned}$$

And so Y_i can be written as a regression equation in terms of the observed factors and error terms.

$$\begin{aligned}
Y_i &= \mu_0(0; \boldsymbol{X}_i) \\
&\quad + D_i \left[\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i) \right] \\
&\quad + Z_i \left[\mu_0(1; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i) \right] \\
&\quad + Z_i D_i \left[\mu_1(1; \boldsymbol{X}_i) - \mu_0(1; \boldsymbol{X}_i) - (\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)) \right] \\
&\quad + (1 - Z_i) \Big[(1 - D_i) U_i(0,0) + D_i U_i(0,1) \Big] \\
&\quad + Z_i \Big[(1 - D_i) U_i(1,0) + D_i U_i(1,1) \Big] \\
&= \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\boldsymbol{X}_i) + \varepsilon_i
\end{aligned}$$

With the following definitions:

(a) $\alpha = \mathbb{E}\left[\mu_0(0; \boldsymbol{X}_i)\right]$ and $\varphi(\boldsymbol{X}_i) = \mu_0(0; \boldsymbol{X}_i) - \alpha$ are the intercept terms.

(b) $\beta = \mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)$ is the indirect effect under $Z_i = 0$

(c) $\gamma = \mu_0(1; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)$ is the direct effect under $D_i = 0$.

(d) $\delta = \mu_1(1; \boldsymbol{X}_i) - \mu_0(1; \boldsymbol{X}_i) - (\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i))$ is the interaction effect.

(e) ε_i is the disruptive error term, with the following definition

$$\varepsilon_i = \begin{cases} (1 - D_i) U_i(0,0) + D_i U_i(0,1), & \text{if } Z_i = 0 \\ (1 - D_i) U_i(1,0) + D_i U_i(1,1), & \text{if } Z_i = 1. \end{cases}$$

Taking the conditional expectation, and collecting for the expressions of the ADE and AIE:

$$\begin{aligned}\text{ADE(z)} = \mathbb{E}\left[Y_i(1, D_i(z)) - Y_i(0, D_i(z))\right] &= \mathbb{E}\left[\gamma + \delta D_i \,|\, Z_i = z\right] \\ \text{AIE(z)} = \mathbb{E}\left[Y_i(z, D_i(1)) - Y_i(z, D_i(0))\right] &= \mathbb{E}\left[\overline{\pi}\left(\beta + \delta z + \widetilde{U}_i(z)\right)\right]\end{aligned}$$

where $\widetilde{U}_i(z) = \mathbb{E}\left[U_i(z,1) - U_i(z,0) \,|\, D_i(0) = 0, D_i(1) = 1, \boldsymbol{X}_i\right]$, for each $z = 0, 1$.

These equations have simpler expressions after assuming constant treatment effects in a linear framework. However, unobserved selection-into-D_i based on unobserved costs and benefits only makes sense in the case of heterogeneous treatment effects. This means a constant treatment effect model is inappropriate for causal mediation analyses of economic settings.

The ultimate regression equation leads to the following expected values, with omitted variables bias coming from the correlated error term, ε_i.

$$\begin{aligned}\mathbb{E}\left[Y_i \,|\, Z_i, D_i, \boldsymbol{X}_i\right] \;=\; & \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\boldsymbol{X}_i) \\ & \underbrace{\begin{aligned}&+ (1 - Z_i)\Big[(1 - D_i)\,\mathbb{E}\left[U_i(0,0) \,|\, D_i(0) = 0, \boldsymbol{X}_i\right] + D_i \mathbb{E}\left[U_i(0,1) \,|\, D_i(0) = 1, \boldsymbol{X}_i\right]\Big] \\ &+ Z_i\Big[(1 - D_i)\,\mathbb{E}\left[U_i(1,0) \,|\, D_i(1) = 0, \boldsymbol{X}_i\right] + D_i \mathbb{E}\left[U_i(1,1) \,|\, D_i(1) = 1, \boldsymbol{X}_i\right]\Big]\end{aligned}}_{\text{Omitted variables bias.}}\end{aligned}$$

If D_i was quasi-randomly assigned, then $\mathbb{E}\left[U_i(z,d) \,|\, D_i(z) = d, \boldsymbol{X}_i\right] = 0$; if not, then there terms comprise non-zero omitted variables bias.

These equations have simpler expressions after assuming constant treatment effects in a linear framework; I have avoided this as having compliers, and controlling for observed factors $\boldsymbol{X}_i$ only makes sense in the case of heterogeneous treatment effects. These regression terms are conventionally estimated in a simultaneous regression (Imai et al. 2010). If sequential quasi-random assignment does not hold, then the regression estimates from estimating the mediation equations (without adjusting for the contaminated bias term) suffer from omitted variables bias.

The following collects these omitted variables bias, omitting the conditional on $\boldsymbol{X}_i$ for brevity.

$$\mathbb{E}\left[Y_i \,|\, Z_i = D_i = 0\right] = \alpha + \mathbb{E}\left[U_i(0,0) \,|\, D_i(0) = 0\right]$$
$$\mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = 1\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = 0\right] = \beta + \Big(\mathbb{E}\left[U_i(0,1) \,|\, D_i(0) = 1\right] - \mathbb{E}\left[U_i(0,0) \,|\, D_i(0) = 0\right]\Big)$$
$$\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i = 0\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = 0\right] = \gamma + \Big(\mathbb{E}\left[U_i(1,0) \,|\, D_i(1) = 0\right] - \mathbb{E}\left[U_i(0,0) \,|\, D_i(0) = 0\right]\Big)$$
$$\begin{aligned}\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i = 1\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i = 0\right] \\ -\Big(\mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = 1\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i = 0\right]\Big) = \delta + \Big(\mathbb{E}\left[U_i(1,1) \,|\, D_i(1) = 1\right] - \mathbb{E}\left[U_i(1,0) \,|\, D_i(1) = 0\right]\Big) \\ - \Big(\mathbb{E}\left[U_i(0,1) \,|\, D_i(0) = 1\right] - \mathbb{E}\left[U_i(0,0) \,|\, D_i(0) = 0\right]\Big)\end{aligned}$$

And so the ADE and AIE estimates are contaminated by these bias terms, because each $\mathbb{E}\left[U_i(z,d) \,|\, D_i(z) = d, \boldsymbol{X}_i\right]$ is not necessary equal to zero. Additionally, the AIE estimates refers to gains from the mediator among $D_i(Z_i)$ compliers (not the entire average), so will be biased when not accounting for $\widetilde{U}_i(z)$, too.

A.5 Roy Model and Sequential quasi-random assignment

Proof of Proposition 1. Suppose Z_i is ignorable, and selection-into-D_i follows a Roy model, with the definitions in Section 4. If selection-into-D_i is degenerate on $U_{0,i}, U_{1,i}$, then for each $z = 0, 1$ all u, u' in the range of $U_{1,i} - U_{0,i}$

$$\mathbb{E}\left[D_i \,|\, Z_i = z, \boldsymbol{X}_i, U_{1,i} - U_{0,i} = u\right] = \mathbb{E}\left[D_i \,|\, Z_i = z, \boldsymbol{X}_i, U_{1,i} - U_{0,i} = u'\right].$$

In this case, the control variables $\boldsymbol{X}_i$, randomised treatment Z_i the costs $\mu_c, U_{c,i}$ are the only determinants of selection-into-D_i — and, $U_{0,i}, U_{1,i}$ play no role. This could be achieved by either assuming that unobserved gains are degenerate (the researcher had observed everything in $\boldsymbol{X}_i$), or selection-into-D_i had been disrupted in some fashion (e.g., by a natural experiment design for D_i).

To motivate a contraposition argument, suppose D_i is ignorable conditional on $Z_i, \boldsymbol{X}_i$. For each $z, d = 0, 1$

$$\begin{aligned}
&D_i \perp\!\!\!\perp Y_i(z,d) \;\mid\; \boldsymbol{X}_i, Z_i = z \\
&\implies D_i \perp\!\!\!\perp \mu_d(z; \boldsymbol{X}_i) + U_{d,i} \;\mid\; \boldsymbol{X}_i, Z_i = z \\
&\implies D_i \perp\!\!\!\perp U_{d,i} \;\mid\; \boldsymbol{X}_i, Z_i = z \\
&\implies D_i \perp\!\!\!\perp U_{1,i} - U_{0,i} \;\mid\; \boldsymbol{X}_i, Z_i = z \\
&\implies \mathbb{E}\left[D_i \,|\, U_{1,i} - U_{0,i} = u', \boldsymbol{X}_i, Z_i = z\right] = \mathbb{E}\left[D_i \,|\, \boldsymbol{X}_i, Z_i = z\right] \\
&\qquad \text{for all } u' \text{ in the range of } U_{1,i} - U_{0,i}.
\end{aligned}$$

This final implication is that selection-into-D_i is degenerate on $U_{0,i}, U_{1,i}$. Thus, a contraposition argument has that if selection-into-D_i is non-degenerate on $U_{0,i}, U_{1,i}$, then D_i is not ignorable.

A.6 Monotonicity $\implies$ Selection Model, in a CM Setting.

Proof that (conditional) monotonicity implies a selection model representation in a CM setting. This proof is an applied example of the Vytlacil (2002) equivalence result, now including conditioning covariates $\boldsymbol{X}_i$, and is presented merely as a validation exercise.

Assume condition monotonicity MTE–1 holds, for any treatment values $z < z'$ and any covariate value $\boldsymbol{X}_i = \boldsymbol{x}$.

$$\Pr\left(D_i(z) \geq D_i(z) \,|\, \boldsymbol{x}\right) = 1.$$

For each value of $\boldsymbol{X}_i = \boldsymbol{x}$ and any treatment values $z < z'$, we first define:

- $\mathcal{A} = \{i : D_i(z) = D_i(z') = 1\}$, always-mediators
- $\mathcal{N} = \{i : D_i(z) = D_i(z') = 0\}$, never-mediators
- $\mathcal{C} = \{i : D_i(z) = 0, D_i(z') = 1\}$, mediator-compliers.

For any mediator complier $i \in \mathcal{C}$, partition the set as follows.

- $\mathcal{Z}_1(i) = \{z : D_i(z) = 1\}$, treatment values where i takes the mediator
- $\mathcal{Z}_0(i) = \{z : D_i(z) = 0\}$, treatment values where i doesn't take the mediator.

Note that having binary $Z_i = 0, 1$ reduces this to the simple case of $\mathcal{Z}_0(i) = \{0\}$, and $\mathcal{Z}_1(i) = \{1\}$. The equivalence result holds for continuous values of Z_i, so continue with the more general $\mathcal{Z}_0(i), \mathcal{Z}_1(i)$ notation.

By monotonicity, we have

$$\sup_{z \in \mathcal{Z}_0(i)} \pi(z; \boldsymbol{x}) \leq \inf_{z \in \mathcal{Z}_1(i)} \pi(z; \boldsymbol{x}), \quad \text{for any } i \in \mathcal{C}$$

where $\pi(z; \boldsymbol{x}) = \Pr\left(D_i = 1 \,|\, Z_i = z, \boldsymbol{X}_i = \boldsymbol{x}\right)$ is the mediator propensity score. A simple proof by contradiction verifies this statement (Vytlacil 2002, Lemma 1).

Now we construct V_i as follows:

$$V_i = \begin{cases} 1, & \text{if } i \in \mathcal{N} \\ 0, & \text{if } i \in \mathcal{A} \\ \inf_{z \in \mathcal{Z}_1(i)} \pi(z; \boldsymbol{x}), & \text{if } i \in \mathcal{C}. \end{cases}$$

Define $\psi(z;\boldsymbol{x}) = \pi(z;\boldsymbol{x})$. Then we can represent $D_i(z)$ as a selection model,

$$D_i(z) = \mathbb{1}\left\{\psi(z;\boldsymbol{X}_i) \geq V_i\right\}, \quad \text{for } z = 0, 1.$$

We can verify this works:

- For $i \in \mathcal{A}$: $V_i = 0$ and $\psi(z;\boldsymbol{x}) \geq 0$ for all z, so $D_i(z) = 1$
- For $i \in \mathcal{N}$: $V_i = 1$ and $\psi(z;\boldsymbol{x}) \leq 1$ for all z, with $\psi(z;\boldsymbol{x}) < 1$ for $z \in \mathcal{Z}_0(i)$, so $D_i(z) = 0$ for $z \in \mathcal{Z}_0(i)$
- For $i \in \mathcal{C}$: $V_i = \inf_{z\in\mathcal{Z}_1(i)} \pi(z;\boldsymbol{x})$
  - When $z \in \mathcal{Z}_1(i)$: $\psi(z;\boldsymbol{x}) \geq \inf_{z'\in\mathcal{Z}_1(i)} \pi(z';\boldsymbol{x}) = V_i$, so $D_i(z) = 1$
  - When $z \in \mathcal{Z}_0(i)$: $\psi(z;\boldsymbol{x}) < \inf_{z'\in\mathcal{Z}_1(i)} \pi(z';\boldsymbol{x}) = V_i$, so $D_i(z) = 0$.

Therefore, the construction $D_i(z) = \mathbb{1}\left\{\psi(z;\boldsymbol{X}_i) \geq V_i\right\}$ is a valid representation of the selection process under monotonicity.

This selection model can be transformed to one with a uniform distribution, to get the general selection model of Heckman & Vytlacil (2005). Let $F_V\left(. \,\middle|\, \boldsymbol{X}_i\right)$ be the conditional cumulative density function of V_i given $\boldsymbol{X}_i$. Define

$$\begin{aligned} U_i &= F_V\left(V_i \mid \boldsymbol{X}_i\right) \\ \pi(z;\boldsymbol{X}_i) &= F_V\left(\psi(z;\boldsymbol{X}_i) \mid \boldsymbol{X}_i\right) = \Pr\left(D_i = 1 \,|\, Z_i = z, \boldsymbol{X}_i\right) \end{aligned}$$

We can then equivalently represent the mediator choice as the transformed selection model

$$D_i(z) = \mathbb{1}\left\{\pi(z;\boldsymbol{X}_i) \geq U_i\right\}, \quad \text{for } z = 0, 1$$

where $U_i \,|\, \boldsymbol{X}_i \sim \text{Uniform}(0, 1)$ by the probability integral transformation.

A.7 Mediator MTE Instrument

This subsection states the relevance, exclusion, assignment, overlap, and support conditions imposed on the mediator instrument $\boldsymbol{X}_i^{\text{IV}}$. Write $\boldsymbol{X}_i = (\boldsymbol{X}_i^{\text{IV}}, \boldsymbol{X}_i^{-})$, where $\boldsymbol{X}_i^{\text{IV}}$ denotes the variables excluded from the outcome equation and $\boldsymbol{X}_i^{-}$ contains the remaining controls. For each treatment arm $z = 0, 1$, define the mediator propensity score

$$\pi\left(z;\boldsymbol{x}^{\text{IV}},\boldsymbol{x}^{-}\right) = \Pr\left(D_i = 1 \mid Z_i = z, \boldsymbol{X}_i^{\text{IV}} = \boldsymbol{x}^{\text{IV}}, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right).$$

Define the realised mediator propensity score as

$$P_i = \pi\big(Z_i; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big).$$

For each z and $\boldsymbol{x}^{-}$, let

$$\mathcal{P}_z(\boldsymbol{x}^{-}) = \mathrm{Supp}\left(P_i \mid Z_i = z, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right)$$

denote the IV-guided propensity-score support within treatment arm z.

(1) *Relevance.* Conditional on $Z_i = z$ and $\boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}$, the mediator propensity score is nonconstant in the excluded mediator instrument:

$$\pi\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big) \text{ is nonconstant in } \boldsymbol{x}^{\mathrm{IV}}, \qquad z = 0, 1.$$

Equivalently, the excluded instrument generates at least two distinct mediator propensity scores within each treatment arm used for identification:

$$K_z(\boldsymbol{x}^{-}) \coloneqq \left|\mathcal{P}_z(\boldsymbol{x}^{-})\right| \geq 2.$$

(2) *Exclusion.* Conditional on Z_i and $\boldsymbol{X}_i^{-}$, the excluded mediator instrument affects the outcome only through mediator take-up:

$$Y_i\big(z, d, \boldsymbol{x}^{\mathrm{IV}}\big) = Y_i(z, d),$$

for all $z, d = 0, 1$ and all $\boldsymbol{x}^{\mathrm{IV}}$ in the relevant support.

(3) *Conditional quasi-random assignment.* Conditional on Z_i and $\boldsymbol{X}_i^{-}$, the excluded mediator instrument is independent of the latent mediator-selection rank and the potential outcomes:

$$\boldsymbol{X}_i^{\mathrm{IV}} \perp\!\!\!\perp \left(U_i, \{Y_i(z, d) : z, d \in \{0, 1\}\}\right) \,\big|\, Z_i, \boldsymbol{X}_i^{-}.$$

Equivalently, variation in $\boldsymbol{X}_i^{\mathrm{IV}}$ changes mediator take-up through the propensity score without independently selecting individuals according to their mediator resistance or potential outcomes.

(4) *Common mediator-selection index.* Treatment and the excluded mediator instrument shift mediator take-up through the same scalar latent rank:

$$D_i\big(z, \boldsymbol{x}^{\mathrm{IV}}\big) = \mathbb{1}\left\{U_i \leq \pi\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big)\right\},$$

for $z = 0, 1$ and all $\boldsymbol{x}^{\mathrm{IV}}$ in the relevant support, almost surely conditional on $\boldsymbol{X}_i^{-}$. Thus,

whenever

$$\pi\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big) \leq \pi\big(z'; \boldsymbol{x}^{\mathrm{IV}\prime}, \boldsymbol{x}^{-}\big),$$

the corresponding mediator-treatment sets are nested:

$$D_i\big(z, \boldsymbol{x}^{\mathrm{IV}}\big) \leq D_i\big(z', \boldsymbol{x}^{\mathrm{IV}\prime}\big) \qquad \text{almost surely conditional on } \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}.$$

(5) *Mediator-cell overlap.* At every propensity-score value used to identify the treatment-arm-specific outcome equation, both mediator states occur with positive conditional probability:

$$0 < \Pr\left(D_i = 1 \mid Z_i = z, P_i = p, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right) = p < 1.$$

Consequently, the mediator-status-specific conditional means

$$\mathbb{E}\left[Y_i \,\middle|\, Z_i = z, D_i = d, P_i = p, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right]$$

are observed for each $d = 0, 1$ and each relevant supported value $p \in \mathcal{P}_z(\boldsymbol{x}^{-})$.

(6) *Mediator-complier support.* For each $\boldsymbol{x} = (\boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-})$, define the treatment-induced mediator-complier interval as

$$\mathcal{C}(\boldsymbol{x}) = \left[\pi\big(0; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big), \pi\big(1; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big)\right].$$

Under mediator monotonicity, the endpoints are ordered so that

$$\pi\big(0; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big) \leq \pi\big(1; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big).$$

Each endpoint of $\mathcal{C}(\boldsymbol{x})$ must lie in the domain over which the restricted treatment-arm-specific outcome curve is identified for both $z = 0, 1$. More precisely,

$$\mathcal{C}(\boldsymbol{x}) \subseteq \mathcal{I}_z(\boldsymbol{x}^{-}), \qquad z = 0, 1,$$

where $\mathcal{I}_z(\boldsymbol{x}^{-})$ is an interval containing $\mathcal{P}_z(\boldsymbol{x}^{-})$ over which the restricted CF basis introduced in Appendix A.8 is defined and identified. If both endpoints belong to $\mathcal{P}_z(\boldsymbol{x}^{-})$, the endpoint conditional means are directly identified in observed data. If an endpoint lies between supported propensity-score values, its identification follows from the maintained restricted-CF specification and the full-rank condition stated below.

The first three conditions are the standard relevance, exclusion, and conditional assignment conditions for a valid mediator instrument. The common-index condition ensures that treatment-induced mediator compliers lie on the same latent selection margin traced by the mediator instrument. Mediator-cell overlap ensures that the selected outcome means for both

mediator states contribute identifying information at the relevant propensity-score values. Mediator-complier support requires the restricted outcome curve within each treatment arm to cover the propensity-score interval generated by changing treatment from $Z_i = 0$ to $Z_i = 1$. The conditions do not require identification at infinity or support over the entire unit interval.

A.8 Restricted MTE-CF Condition

Proposition 2 uses mediator-status-specific CFs to identify a restricted conditional-outcome curve separately within the $Z_i = 0$ and $Z_i = 1$ treatment arms. The finite support of the mediator instrument does not nonparametrically identify an unrestricted CF or the derivative of an unrestricted conditional-outcome function. Instead, the paper imposes a finite-dimensional restriction on the CFs and uses the mediator-status-specific outcome means at the supported propensity-score values to identify its coefficients.

For each treatment arm $z = 0, 1$, define the selected CFs

$$g_{z,0}\big(p, \boldsymbol{x}^-\big) = \mathbb{E}\left[U_i(z,0) \,\middle|\, U_i > p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right],$$

$$g_{z,1}\big(p, \boldsymbol{x}^-\big) = \mathbb{E}\left[U_i(z,1) \,\middle|\, U_i \leq p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

These CFs are allowed to differ across treatment arms because the relevant potential-outcome residuals are indexed jointly by z and d.

The treatment-arm-specific mediator-status outcome means are

$$\begin{aligned}\mathbb{E}\left[Y_i \,\middle|\, Z_i = z, D_i = 0, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right] \\ = \mu_0\big(z; \boldsymbol{x}^-\big) + g_{z,0}\big(p, \boldsymbol{x}^-\big),\end{aligned}$$

$$\begin{aligned}\mathbb{E}\left[Y_i \,\middle|\, Z_i = z, D_i = 1, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right] \\ = \mu_1\big(z; \boldsymbol{x}^-\big) + g_{z,1}\big(p, \boldsymbol{x}^-\big).\end{aligned}$$

Equivalently, the pooled CF-augmented estimating equation is

$$\begin{aligned}Y_i = \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi\big(\boldsymbol{X}_i^-\big) \\ + (1 - Z_i)(1 - D_i) g_{0,0}(P_i) + (1 - Z_i) D_i g_{0,1}(P_i) \\ + Z_i(1 - D_i) g_{1,0}(P_i) + Z_i D_i g_{1,1}(P_i) + \epsilon_i,\end{aligned}$$

where

$$\mathbb{E}\left[\epsilon_i \,\middle|\, Z_i, D_i, P_i, \boldsymbol{X}_i^-\right] = 0.$$

This equation is an estimating representation and does not by itself require each of $\alpha, \beta, \gamma, \delta$ to be separately interpreted as an identified CM effect.

A.8.1 Separability from observed heterogeneity

The first restriction assumes that, after observed heterogeneity has been absorbed by the potential-outcome means, the shape of selection on latent mediator resistance is common across values of $\boldsymbol{X}_i^-$. For each $z, d = 0, 1$, define the rank-specific residual mean

$$r_{z,d}\big(u, \boldsymbol{x}^-\big) = \mathbb{E}\left[U_i(z,d) \,\middle|\, U_i = u, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

The separability restriction is

$$r_{z,d}\big(u, \boldsymbol{x}^-\big) = r_{z,d}(u), \qquad z, d = 0, 1.$$

Equivalently, $\boldsymbol{X}_i^-$ may shift the observed potential-outcome means, but it does not unrestrictedly alter the shape of selection on the unobserved mediator rank. A linear empirical specification for the observed heterogeneity may be written as

$$\mu_d\big(z; \boldsymbol{x}^-\big) = \alpha + \beta d + \gamma z + \delta z d + \boldsymbol{\phi}_{z,d}' \boldsymbol{x}^-.$$

A.8.2 Finite arm-specific propensity-score support

For each z and $\boldsymbol{x}^-$, write

$$\mathcal{P}_z(\boldsymbol{x}^-) = \left\{p_{z1}(\boldsymbol{x}^-), \ldots, p_{zK_z}(\boldsymbol{x}^-)\right\},$$

where

$$K_z(\boldsymbol{x}^-) = \left|\mathcal{P}_z(\boldsymbol{x}^-)\right|.$$

The number $K_z(\boldsymbol{x}^-)$ counts the distinct propensity-score values generated by $\boldsymbol{X}_i^{\mathrm{IV}}$ within treatment arm z, rather than the number of raw values taken by the instrument. The propensity-score support and the CFs are allowed to differ between the $Z_i = 0$ and $Z_i = 1$ arms.

At every supported propensity score, observed data identify the two selected conditional-outcome means

$$h_{z,0}\big(p_{zk}, \boldsymbol{x}^-\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = z, D_i = 0, P_i = p_{zk}, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right],$$

and

$$h_{z,1}\big(p_{zk}, \boldsymbol{x}^-\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = z, D_i = 1, P_i = p_{zk}, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

The mediator-cell overlap condition in Appendix A.7 ensures that both conditional means are observed at each support point used for identification.

A.8.3 Restricted CF basis

For each $z, d = 0, 1$, let

$$\boldsymbol{b}_{z,d}(p) = \big(b_{z,d,1}(p), \ldots, b_{z,d,L_{z,d}}(p)\big)'$$

denote a vector of known basis functions. The restricted CF specification is

$$g_{z,d}(p) = c_{z,d} + \boldsymbol{b}_{z,d}(p)'\boldsymbol{\rho}_{z,d}, \qquad p \in \mathcal{I}_z(\boldsymbol{x}^-).$$

The constants $c_{z,d}$ are not separately distinguished from the corresponding potential-outcome means. Accordingly, impose the normalisation

$$c_{z,d} = 0, \qquad z, d = 0, 1,$$

or equivalently omit the constant from each CF basis and absorb it into $\mu_d(z; \boldsymbol{x}^-)$. The number of free basis coefficients must not exceed the identifying variation supplied by the observed mediator-status-specific conditional means.

The basis functions must also be compatible with the truncated-mean interpretation of the CFs. In particular, the rank-specific residual means satisfy

$$\int_0^1 r_{z,d}(u)\, du = 0, \qquad z, d = 0, 1,$$

and the relevant selected residual means are generated by integrating these rank-specific residual means over the selected regions. For example,

$$\mathbb{E}\left[U_i(z, d) \,|\, U_i \leq p\right] = \frac{1}{p}\int_0^p r_{z,d}(u)\, du,$$

and

$$\mathbb{E}\left[U_i(z, d) \,|\, U_i > p\right] = \frac{1}{1-p}\int_p^1 r_{z,d}(u)\, du.$$

The finite-dimensional basis must preserve these mean-zero and truncated-mean restrictions. This compatibility requirement prevents the CFs from being arbitrary unrelated functions of the propensity score.

A.8.4 Full-rank condition

Collect the free potential-outcome mean coefficients and the CF basis coefficients within treatment arm z in the vector

$$\boldsymbol{\vartheta}_z = \left(\boldsymbol{\theta}'_{z,0}, \boldsymbol{\theta}'_{z,1}, \boldsymbol{\rho}'_{z,0}, \boldsymbol{\rho}'_{z,1}\right)',$$

where $\boldsymbol{\theta}_{z,d}$ parameterises the observed component of $\mu_d(z; \boldsymbol{X}_i^-)$. Stack the identified mediator-status-specific conditional means at the supported propensity-score values in

$$\boldsymbol{h}_z(\boldsymbol{x}^-) = \big(h_{z,0}(p_{z1}, \boldsymbol{x}^-), \ldots, h_{z,0}(p_{zK_z}, \boldsymbol{x}^-), h_{z,1}(p_{z1}, \boldsymbol{x}^-), \ldots, h_{z,1}(p_{zK_z}, \boldsymbol{x}^-)\big)'.$$

The restricted outcome equations imply the finite system

$$\boldsymbol{h}_z(\boldsymbol{x}^-) = \mathbf{Q}_z(\boldsymbol{x}^-)\boldsymbol{\vartheta}_z,$$

where $\mathbf{Q}_z(\boldsymbol{x}^-)$ contains the observed outcome regressors and the evaluations of the known CF bases at the supported propensity-score values. Identification requires

$$\operatorname{rank}\left[\mathbf{Q}_z(\boldsymbol{x}^-)\right] = \dim\left(\boldsymbol{\vartheta}_z\right), \qquad z = 0, 1.$$

The rank condition requires the supported propensity-score values to generate linearly independent evaluations of the maintained CF basis after accounting for the observed outcome regressors and the imposed normalisations. Dimension counting alone is insufficient, because distinct basis coefficients may otherwise generate the same conditional means at every supported propensity-score value.

A.8.5 Restricted conditional-outcome curve

The conditional outcome mean that averages over realised mediator status is

$$m\big(p; z, \boldsymbol{x}^-\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = z, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

Because

$$\mathbb{E}\left[D_i \,\middle|\, Z_i = z, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right] = p,$$

the CF-augmented estimating equation implies

$$\begin{aligned} m^R\big(p; z, \boldsymbol{x}^-\big) = (1-p)\mu_0\big(z; \boldsymbol{x}^-\big) + p\mu_1\big(z; \boldsymbol{x}^-\big) \\ + (1-p)g_{z,0}(p) + pg_{z,1}(p). \end{aligned}$$

At every observed support point,

$$m^R\big(p_{zk}; z, \boldsymbol{x}^-\big) = m\big(p_{zk}; z, \boldsymbol{x}^-\big).$$

The restricted-CF specification and full-rank condition identify the parameters of m^R from the observed conditional means. They therefore identify the restricted extension $m^R(p; z, \boldsymbol{x}^-)$ over the interval $\mathcal{I}_z(\boldsymbol{x}^-)$. Values of the outcome curve between finite support points are

identified within this maintained finite-dimensional class, rather than by nonparametric local differentiation.

A.9 MTE Identification of the ADE and AIE

Proof of Theorem MTE Approach. Assume that Assumptions MTE–1, MTE–2, MTE–3, and MTE–CF hold.

Proposition 2 establishes identification of the restricted conditional-outcome curve separately within each treatment arm:

$$m^R(p; z, \boldsymbol{x}^-) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = z, P_i = p, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right],$$

for $z = 0, 1$ and for p in the interval $\mathcal{I}_z(\boldsymbol{x}^-)$ covered by the restricted CF model.

The superscript R indicates that values of the curve between finitely supported propensity-score values are identified within the maintained restricted-CF class. At observed propensity-score support points, m^R coincides with the observed conditional-outcome mean. At other points in $\mathcal{I}_z(\boldsymbol{x}^-)$, m^R is identified by the finite-dimensional CF restriction and the full-column-rank condition.

A.9.1 Interpretation of the treatment-arm-specific outcome curve

Under the common mediator-selection index,

$$D_i(z', \boldsymbol{x}^{\mathrm{IV}}) = \mathbb{1}\left\{U_i \le \pi(z'; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{X}_i^-)\right\}.$$

For any fixed treatment state z and propensity-score value p, define the counterfactual mediator rule

$$D_i(p) = \mathbb{1}\left\{U_i \le p\right\}.$$

Conditional exclusion and quasi-random assignment of the mediator instrument imply

$$m^R(p; z, \boldsymbol{x}^-) = \mathbb{E}\left[Y_i\big(z, D_i(p)\big) \,\middle|\, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right].$$

Accordingly, evaluating the restricted outcome curve at the treatment-induced propensity score gives

$$\begin{aligned} &m^R\left(\pi(z'; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^-); z, \boldsymbol{x}^-\right) \\ &\quad = \mathbb{E}\left[Y_i\left(z, D_i(z', \boldsymbol{x}^{\mathrm{IV}})\right) \,\middle|\, \boldsymbol{X}_i^- = \boldsymbol{x}^-\right], \end{aligned}$$

for $z, z' = 0, 1$.

Therefore, the two endpoint values of the $Z_i = z$ outcome curve identify the two relevant

nested counterfactual outcome means:

$$m^R\left(\pi\left(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right);z,\boldsymbol{x}^{-}\right) = \mathbb{E}\left[Y_i\left(z,D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\right)\,\middle|\,\boldsymbol{X}_i^{-}=\boldsymbol{x}^{-}\right],$$

and

$$m^R\left(\pi\left(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right);z,\boldsymbol{x}^{-}\right) = \mathbb{E}\left[Y_i\left(z,D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right)\right)\,\middle|\,\boldsymbol{X}_i^{-}=\boldsymbol{x}^{-}\right].$$

A.9.2 Identification of the AIE

For $z=0,1$, define the conditional indirect effect as

$$\mathrm{AIE}\left(z;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right) = \mathbb{E}\left[Y_i\left(z,D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right)\right) - Y_i\left(z,D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\right)\,\middle|\,\boldsymbol{X}_i^{-}=\boldsymbol{x}^{-}\right].$$

Using (A.9.1),

$$\begin{aligned}\mathrm{AIE}\left(z;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right) &= m^R\left(\pi\left(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right);z,\boldsymbol{x}^{-}\right)\\ &\quad - m^R\left(\pi\left(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right);z,\boldsymbol{x}^{-}\right).\end{aligned}$$

The mediator-complier support condition ensures that both propensity-score endpoints lie in the domain $\mathcal{I}_z(\boldsymbol{x}^{-})$ over which m^R is identified.

Therefore, the right-hand side of (A.9.2) is identified for each $z=0,1$ and each relevant covariate value.

To connect this endpoint expression to the mediator MTE, note that mediator monotonicity and the common-index representation imply

$$\begin{aligned}D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right) - D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\\ = \mathbb{1}\left\{\pi\left(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right) < U_i \le \pi\left(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right)\right\}.\end{aligned}$$

For binary mediator values,

$$\begin{aligned}Y_i\left(z,D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right)\right) - Y_i\left(z,D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\right)\\ = \left[D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right) - D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\right]\left[Y_i(z,1) - Y_i(z,0)\right].\end{aligned}$$

Substituting (A.9.2) gives

$$\begin{aligned}Y_i\left(z,D_i\left(1,\boldsymbol{x}^{\mathrm{IV}}\right)\right) - Y_i\left(z,D_i\left(0,\boldsymbol{x}^{\mathrm{IV}}\right)\right)\\ = \mathbb{1}\left\{\pi\left(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right) < U_i \le \pi\left(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right)\right\}\left[Y_i(z,1) - Y_i(z,0)\right].\end{aligned}$$

Taking conditional expectations and iterating over the latent mediator rank yields

$$\begin{aligned}&\mathrm{AIE}\left(z;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right)\\&\quad=\int_{\pi(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}^{\pi(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}\mathbb{E}\left[Y_i(z,1)-Y_i(z,0)\,\middle|\,U_i=p,\boldsymbol{X}_i^{-}=\boldsymbol{x}^{-}\right]\,dp.\end{aligned}$$

By the definition of the mediator MTE,

$$\begin{aligned}&\mathrm{AIE}\left(z;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\right)\\&\quad=\int_{\pi(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}^{\pi(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}\mathrm{MTE}\left(p;z,\boldsymbol{x}^{-}\right)\,dp.\end{aligned}$$

Within the restricted-CF model, the identified mediator MTE is the restricted mediator MTE,

$$\mathrm{MTE}^{R}\left(p;z,\boldsymbol{x}^{-}\right)=\frac{\partial}{\partial p}m^{R}\big(p;z,\boldsymbol{x}^{-}\big),$$

for almost every p in $\mathcal{I}_z(\boldsymbol{x}^{-})$.

Absolute continuity of m^R therefore gives

$$\begin{aligned}&\int_{\pi(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}^{\pi(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-})}\mathrm{MTE}^{R}\left(p;z,\boldsymbol{x}^{-}\right)\,dp\\&\quad=m^{R}\left(\pi\big(1;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\big);z,\boldsymbol{x}^{-}\right)\\&\qquad-m^{R}\left(\pi\big(0;\boldsymbol{x}^{\mathrm{IV}},\boldsymbol{x}^{-}\big);z,\boldsymbol{x}^{-}\right).\end{aligned}$$

Thus, the MTE integral and endpoint-difference representations identify the same conditional indirect effect within the maintained restricted model.

Averaging over the joint distribution of $\boldsymbol{X}_i^{\mathrm{IV}}$ and $\boldsymbol{X}_i^{-}$ identifies

$$\begin{aligned}\mathrm{AIE}(z)=\mathbb{E}\Big[&m^{R}\left(\pi\big(1;\boldsymbol{X}_i^{\mathrm{IV}},\boldsymbol{X}_i^{-}\big);z,\boldsymbol{X}_i^{-}\right)\\&-m^{R}\left(\pi\big(0;\boldsymbol{X}_i^{\mathrm{IV}},\boldsymbol{X}_i^{-}\big);z,\boldsymbol{X}_i^{-}\right)\Big].\end{aligned}$$

Equivalently,

$$\mathrm{AIE}(z)=\mathbb{E}\left[\int_{\pi(0;\boldsymbol{X}_i^{\mathrm{IV}},\boldsymbol{X}_i^{-})}^{\pi(1;\boldsymbol{X}_i^{\mathrm{IV}},\boldsymbol{X}_i^{-})}\mathrm{MTE}^{R}\left(u;z,\boldsymbol{X}_i^{-}\right)\,du\right].$$

This establishes identification of $\mathrm{AIE}(z)$ for each $z=0,1$.

A.9.3 Identification of the ADE

For $z = 0, 1$, recall the ADE definition,

$$\text{ADE}(z) = \mathbb{E}\left[Y_i\big(1, D_i(z)\big) - Y_i\big(0, D_i(z)\big)\right].$$

The total treatment effect is

$$\text{ATE} = \mathbb{E}\left[Y_i\big(1, D_i(1)\big) - Y_i\big(0, D_i(0)\big)\right].$$

First add and subtract $Y_i(0, D_i(1))$:

$$\begin{aligned}\text{ATE} = \mathbb{E}&\left[Y_i\big(1, D_i(1)\big) - Y_i\big(0, D_i(1)\big)\right] \\ &+ \mathbb{E}\left[Y_i\big(0, D_i(1)\big) - Y_i\big(0, D_i(0)\big)\right].\end{aligned}$$

The first term is ADE(1) and the second is AIE(0).

Therefore,

$$\text{ATE} = \text{ADE}(1) + \text{AIE}(0).$$

It follows that

$$\text{ADE}(1) = \text{ATE} - \text{AIE}(0).$$

Alternatively, AIE(1) and ADE(0) are identified by the same process, identifying AIE(1) via the Mediator MTE among the $Z_i = 1$ sample and using ADE(0) as the ATE residual. This proves Theorem MTE Approach.

Below is the complete appendix subsection, organised around estimation of the CM effects. It treats estimation of the propensity score and restricted outcome curve as intermediate steps, then states consistency, asymptotic normality, and bootstrap inference. The cited literature supports series/spline consistency and asymptotic normality of smooth functionals, and the adjustment required when the propensity score is a generated regressor.

A.10 Semi-parametric Estimation of the AIE and ADE

This subsection describes the semi-parametric estimation of the AIE and ADE. For notational convenience, the exposition focuses on the $Z_i = 0$ treatment arm, which identifies AIE(0) and, together with the ATE, ADE(1). The estimators of AIE(1) and ADE(0) follow symmetrically by applying the same procedure to the $Z_i = 1$ treatment arm.

Let

$$n_0 \coloneqq \sum_{i=1}^{n} \mathbb{1}\left\{Z_i = 0\right\}$$

denote the number of observations assigned to the $Z_i = 0$ arm.

A.10.1 Estimation of the Mediator Propensity Score

The first stage estimates the mediator propensity score,

$$\pi\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big) = \Pr\left(D_i = 1 \,\middle|\, Z_i = z, \boldsymbol{X}_i^{\mathrm{IV}} = \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right).$$

Let

$$\widehat{\pi}\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^{-}\big)$$

denote its estimate, and define the estimated realised propensity score as

$$\widehat{P}_i = \widehat{\pi}\big(Z_i; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big).$$

The propensity score may be estimated using a correctly specified binary-response model or an appropriate semi-parametric estimator. The estimator must consistently estimate both the realised propensity score and the two counterfactual endpoint propensities,

$$\widehat{p}_{0i} \coloneqq \widehat{\pi}\big(0; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big)$$

and

$$\widehat{p}_{1i} \coloneqq \widehat{\pi}\big(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big).$$

The mediator-complier support condition requires both endpoint propensities to lie in the domain identified by mediator-instrument variation within the treatment arm used to estimate the outcome curve. For the $Z_i = 0$ estimator, this requires

$$\big[\pi\big(0; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big), \pi\big(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big)\big] \subseteq \mathcal{I}_0\big(\boldsymbol{X}_i^{-}\big)$$

almost surely, where $\mathcal{I}_0(\boldsymbol{X}_i^{-})$ is the domain over which the restricted $Z_i = 0$ outcome curve is identified.

A.10.2 Estimation of the Restricted Outcome Curve

Within the $Z_i = 0$ treatment arm, define

$$m\big(p; 0, \boldsymbol{x}^{-}\big) = \mathbb{E}\left[Y_i \,\middle|\, Z_i = 0, P_i = p, \boldsymbol{X}_i^{-} = \boldsymbol{x}^{-}\right].$$

Let

$$\boldsymbol{b}_{0,K}(p) = \big(b_{01}(p), \ldots, b_{0K}(p)\big)'$$

be a vector of known polynomial, regression-spline, or other series basis functions. The

restricted outcome curve is approximated by

$$m_K^R\left(p;0,\boldsymbol{x}^-\right)=\boldsymbol{\phi}_0'\boldsymbol{x}^-+\boldsymbol{b}_{0,K}(p)'\boldsymbol{\rho}_0.$$

An intercept is included either in $\boldsymbol{x}^-$ or in $\boldsymbol{b}_{0,K}(p)$, but not in both. More generally, the basis must not duplicate terms already included in the linear control function. The rank condition in Appendix A.8 requires the IV-guided propensity-score variation to distinguish the maintained basis terms from one another and from the included controls.

The coefficients are estimated in the $Z_i=0$ subsample by

$$\left(\widehat{\boldsymbol{\phi}}_0,\widehat{\boldsymbol{\rho}}_0\right)=\arg\min_{\boldsymbol{\phi},\boldsymbol{\rho}}\frac{1}{n_0}\sum_{i:Z_i=0}\left[Y_i-\boldsymbol{\phi}'\boldsymbol{X}_i^-\;-\;\boldsymbol{b}_{0,K}\left(\widehat{P}_i\right)'\boldsymbol{\rho}\right]^2.$$

The resulting fitted outcome curve is

$$\widehat{m}\left(p;0,\boldsymbol{x}^-\right)=\widehat{\boldsymbol{\phi}}_0'\boldsymbol{x}^-+\boldsymbol{b}_{0,K}(p)'\widehat{\boldsymbol{\rho}}_0.$$

When the maintained restricted outcome model is exactly finite dimensional, K is fixed. When the basis is used as a sieve approximation to a smooth outcome curve, $K=K_{n_0}$ may increase with the treatment-arm sample size. The basis dimension must increase sufficiently quickly to control approximation error, but sufficiently slowly to control estimation error (Newey 1997, Chen 2007).

A parametric assumption on the joint distribution of the unobserved mediator-selection and outcome errors provides an alternative specification of the propensity-score component of the outcome curve. For example, joint normality implies the familiar inverse-Mills-ratio correction. The semi-parametric estimator instead uses a polynomial, spline, or sieve approximation and does not require a parametric distribution for the unobserved errors.

A.10.3 Estimation of the AIE and ADE

For each observation in the $Z_i=0$ subsample, evaluate the fitted $Z_i=0$ outcome curve at the mediator propensities induced by both treatment states:

$$\widehat{m}\left(\widehat{p}_{0i};0,\boldsymbol{X}_i^-\right)$$

and

$$\widehat{m}\left(\widehat{p}_{1i};0,\boldsymbol{X}_i^-\right).$$

The fitted conditional indirect effect is

$$\widehat{\tau}_i^{\mathrm{AIE}(0)} = \widehat{m}\big(\widehat{p}_{1i}; 0, \boldsymbol{X}_i^-\big) - \widehat{m}\big(\widehat{p}_{0i}; 0, \boldsymbol{X}_i^-\big).$$

Under the partially linear specification in (A.10.2), the additive control-variable component cancels:

$$\widehat{\tau}_i^{\mathrm{AIE}(0)} = \big[\boldsymbol{b}_{0,K}\big(\widehat{p}_{1i}\big) - \boldsymbol{b}_{0,K}\big(\widehat{p}_{0i}\big)\big]' \widehat{\boldsymbol{\rho}}_0.$$

The cancellation of the additive control term does not make the controls redundant. They remain necessary for conditional instrument validity, consistent propensity-score estimation, and the separability condition used to identify the restricted outcome curve.

The indirect effect is estimated by averaging these fitted endpoint differences over the $Z_i = 0$ subsample:

$$\boxed{\widehat{\mathrm{AIE}}_0(0) = \frac{1}{n_0} \sum_{i:Z_i=0} \widehat{\tau}_i^{\mathrm{AIE}(0)}}.$$

Equivalently,

$$\widehat{\mathrm{AIE}}_0(0) = \frac{1}{n_0} \sum_{i:Z_i=0} \Big[\widehat{m}\big(\widehat{p}_{1i}; 0, \boldsymbol{X}_i^-\big) - \widehat{m}\big(\widehat{p}_{0i}; 0, \boldsymbol{X}_i^-\big)\Big].$$

Quasi-random assignment of Z_i implies that the distribution of

$$\big(\boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^-\big)$$

within the $Z_i = 0$ subsample consistently estimates the corresponding target-population distribution. The subsample average therefore standardises the conditional endpoint differences to the target population.

An alternative estimator standardises the fitted effects over the full sample:

$$\widehat{\mathrm{AIE}}_{\mathrm{std}}(0) = \frac{1}{n} \sum_{i=1}^{n} \Big[\widehat{m}\big(\widehat{p}_{1i}; 0, \boldsymbol{X}_i^-\big) - \widehat{m}\big(\widehat{p}_{0i}; 0, \boldsymbol{X}_i^-\big)\Big].$$

Under quasi-random treatment assignment, the subsample and full-sample standardisation estimators have the same probability limit, although their finite-sample and asymptotic variances may differ.

The complementary direct effect is estimated using

$$\mathrm{ATE} = \mathrm{AIE}(0) + \mathrm{ADE}(1).$$

Thus,

$$\boxed{\widehat{\mathrm{ADE}}(1) = \widehat{\mathrm{ATE}} - \widehat{\mathrm{AIE}}(0)}.$$

Repeating the outcome-curve estimation in the $Z_i = 1$ subsample gives

$$\widehat{\mathrm{AIE}}(1)$$

and

$$\boxed{\widehat{\mathrm{ADE}}(0) = \widehat{\mathrm{ATE}} - \widehat{\mathrm{AIE}}(1)}.$$

The endpoint estimator does not require numerical differentiation. Under absolute continuity,

$$\begin{aligned} &m^R\big(p_1; 0, \boldsymbol{x}^-\big) - m^R\big(p_0; 0, \boldsymbol{x}^-\big) \\ &\quad = \int_{p_0}^{p_1} \mathrm{MTE}^R\left(u; 0, \boldsymbol{x}^-\right) du. \end{aligned}$$

The AIE can therefore be estimated directly from the two fitted endpoint values. This avoids the instability associated with numerically differentiating a flexible estimate of the conditional-outcome curve.

Prediction at

$$\widehat{\pi}\big(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^-\big)$$

is not extrapolation merely because the outcome curve is estimated in the $Z_i = 0$ subsample. It is interpolation when both treatment-induced propensity scores lie in the domain $\mathcal{I}_0(\boldsymbol{X}_i^-)$ identified by mediator-instrument variation and the restricted-CF model within the $Z_i = 0$ arm. If an endpoint lies outside that domain, its evaluation instead requires unsupported extrapolation.

A.10.4 Consistency and Asymptotic Normality

The large-sample argument combines standard results for series or sieve estimation with results for multi-step semi-parametric estimators containing generated regressors. Series estimators consistently approximate conditional expectations under smoothness, moment, support, approximation, and basis-growth conditions. Although the estimated outcome curve may converge at a nonparametric rate, sufficiently smooth scalar functionals of the curve can be root-n consistent and asymptotically normal (Newey 1997, Chen 2007).

Assume that

$$q_0 \coloneqq \Pr\left(Z_i = 0\right)$$

satisfies

$$0 < q_0 < 1.$$

Then

$$\frac{n_0}{n} \xrightarrow{p} q_0.$$

In addition to the identification assumptions, impose the following regularity conditions:

(1) The observations are independently and identically distributed, or satisfy an appropriate weak-dependence condition.

(2) The outcome has sufficiently many finite moments. For some $\epsilon > 0$,

$$\mathbb{E}\left[|Y_i|^{2+\epsilon}\right] < \infty.$$

(3) The mediator propensities are bounded away from zero and one over the target population:

$$0 < \underline{p} \leq \pi\big(z; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big) \leq \overline{p} < 1, \qquad z = 0, 1.$$

(4) Both treatment-induced propensity-score endpoints lie in the domain of the restricted outcome curve:

$$\big[\pi\big(0; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big), \pi\big(1; \boldsymbol{X}_i^{\mathrm{IV}}, \boldsymbol{X}_i^{-}\big)\big] \subseteq \mathcal{I}_0\big(\boldsymbol{X}_i^{-}\big)$$

almost surely.

(5) The population second-moment matrix of the outcome-curve regressors is nonsingular, with its relevant eigenvalues bounded away from zero and infinity.

(6) The restricted outcome curve is sufficiently smooth and can be approximated by the chosen basis with error

$$a_K \coloneqq \sup_{p, \boldsymbol{x}^-} \left| m^R\big(p; 0, \boldsymbol{x}^-\big) - m_K^R\big(p; 0, \boldsymbol{x}^-\big) \right| \longrightarrow 0.$$

(7) If the sieve dimension increases, then

$$K = K_{n_0} \longrightarrow \infty$$

sufficiently slowly for sampling error to vanish and sufficiently quickly for the approximation bias in the endpoint functional to be asymptotically negligible.

(8) The first-stage estimator is uniformly consistent over the relevant support:

$$\sup_{z, \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^-} \left| \widehat{\pi}\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^-\big) - \pi\big(z; \boldsymbol{x}^{\mathrm{IV}}, \boldsymbol{x}^-\big) \right| \xrightarrow{p} 0.$$

(9) For asymptotic normality, the first-stage estimator possesses an asymptotically linear representation, and the map from the first-stage propensity score and restricted outcome curve to the averaged endpoint difference is sufficiently smooth.

Under these conditions,

$$\sup_{p,\boldsymbol{x}^-} \left|\widehat{m}\big(p;0,\boldsymbol{x}^-\big) - m^R\big(p;0,\boldsymbol{x}^-\big)\right| \xrightarrow{p} 0$$

over the mediator-complier domain.

Uniform consistency of the fitted curve and consistency of the estimated endpoint propensities imply

$$\begin{aligned}\widehat{m}\big(\widehat{p}_{1i};0,\boldsymbol{X}_i^-\big) - \widehat{m}\big(\widehat{p}_{0i};0,\boldsymbol{X}_i^-\big)\\ - \big[m^R\big(p_{1i};0,\boldsymbol{X}_i^-\big) - m^R\big(p_{0i};0,\boldsymbol{X}_i^-\big)\big] \xrightarrow{p} 0.\end{aligned}$$

The law of large numbers therefore gives

$$\widehat{\mathrm{AIE}}_0(0) \xrightarrow{p} \mathrm{AIE}(0).$$

If

$$\widehat{\mathrm{ATE}} \xrightarrow{p} \mathrm{ATE},$$

then

$$\widehat{\mathrm{ADE}}(1) \xrightarrow{p} \mathrm{ADE}(1).$$

The entire estimated outcome curve need not attain a parametric convergence rate. The AIE, however, is a scalar functional of the propensity-score function, the restricted outcome curve, and the covariate distribution:

$$\begin{aligned}\mathcal{T}_0\big(m,\pi,F_0\big) = \int \Big[m\big(\pi(1;\boldsymbol{x});0,\boldsymbol{x}^-\big)\\ - m\big(\pi(0;\boldsymbol{x});0,\boldsymbol{x}^-\big)\Big] dF_0(\boldsymbol{x}),\end{aligned}$$

where F_0 is the covariate distribution in the $Z_i = 0$ arm.

Under sufficient smoothness, a negligible sieve-approximation bias, and appropriate regularity of the generated propensity score, this functional admits an asymptotically linear representation:

$$\widehat{\mathrm{AIE}}_0(0) - \mathrm{AIE}(0) = \frac{1}{n_0}\sum_{i:Z_i=0}\psi_{0i} + o_p\left(n_0^{-1/2}\right).$$

The influence function ψ_{0i} contains contributions from:

(i) sampling variation in the covariate distribution;

(ii) estimation of the restricted outcome curve;

(iii) estimation of the propensity score used as a generated regressor;

(iv) estimation of the two endpoint propensity scores.

The estimated propensity score has a dual role: it is both a conditioning variable in the restricted outcome regression and the argument at which the fitted outcome curve is evaluated. The contribution of both roles must be included in the asymptotic variance (Hahn & Ridder 2013).

It follows that

$$\sqrt{n_0}\left[\widehat{\mathrm{AIE}}_0(0) - \mathrm{AIE}(0)\right] \xrightarrow{d} \mathcal{N}\left(0, V_{\mathrm{AIE}(0)}\right),$$

where

$$V_{\mathrm{AIE}(0)} = \mathbb{E}\left[\psi_{0i}^2 \,\middle|\, Z_i = 0\right].$$

Because

$$\frac{n_0}{n} \xrightarrow{p} q_0,$$

the equivalent full-sample normalisation is

$$\sqrt{n}\left[\widehat{\mathrm{AIE}}_0(0) - \mathrm{AIE}(0)\right] \xrightarrow{d} \mathcal{N}\left(0, \frac{V_{\mathrm{AIE}(0)}}{q_0}\right).$$

Thus, use of the $Z_i = 0$ subsample affects the asymptotic variance through the treatment-arm probability but does not change the convergence order when q_0 is bounded away from zero.

If the ATE estimator is also asymptotically linear, then

$$\begin{aligned}\widehat{\mathrm{ADE}}(1) - \mathrm{ADE}(1) = &\left[\widehat{\mathrm{ATE}} - \mathrm{ATE}\right] \\ &- \left[\widehat{\mathrm{AIE}}(0) - \mathrm{AIE}(0)\right].\end{aligned}$$

The ADE estimator is therefore asymptotically normal, with variance accounting for the sampling variance and covariance of the estimated ATE and AIE.

A.10.5 Bootstrap Inference

Inference is implemented using a nonparametric bootstrap that repeats the complete estimation procedure. For each bootstrap replication:

(1) Resample observations from the original sample.

(2) Re-estimate the mediator propensity-score model.

(3) Reconstruct the realised propensity scores and the treatment-induced endpoint propensities.

(4) Re-estimate the restricted conditional-outcome curve using the bootstrap $Z_i = 0$ subsample.

(5) Evaluate the fitted outcome curve at both endpoint propensity scores.

(6) Average the endpoint differences to estimate AIE(0).

(7) Re-estimate the ATE.

(8) Construct

$$\widehat{\text{ADE}}(1) = \widehat{\text{ATE}} - \widehat{\text{AIE}}(0).$$

Repeating every stage accounts for uncertainty from propensity-score estimation, estimation of the restricted outcome curve, endpoint evaluation, standardisation over the covariate distribution, and estimation of the ATE. Under the regularity conditions supporting the asymptotic linear representation, the bootstrap consistently estimates the sampling distributions of the AIE and ADE estimators.

The same procedure applies to estimation of the $Z_i = 1$ outcome curve and the resulting estimators of AIE(1) and ADE(0).

A.11 Implementation and Further Simulation Evidence

A number of statistical packages, for the R language (R Core Team 2025), made the simulation analysis for this paper possible.

- *Tidyverse* (Wickham, Averick, Bryan, Chang, McGowan, François, Grolemund, Hayes, Henry, Hester, Kuhn, Pedersen, Miller, Bache, Müller, Ooms, Robinson, Seidel, Spinu, Takahashi, Vaughan, Wilke, Woo & Yutani 2019) collected tools for data analysis in the R language.

- *Mgcv* (Wood, N., Pya & S"afken 2016) allows semi-parametric estimation, using splines, in the R language.

- *Mediate* (Tingley, Yamamoto, Hirose, Keele & Imai 2014) automates the sequential quasi-random assignment estimates of CM effects (Imai et al. 2010) in the R language.